Risk and Return

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1900s – Roots of Randomness

- Louis Bachelier's Theory of Speculation
- First to model asset prices using random walks & Brownian motion
- Risk implied by price volatility

1950s – Modern Portfolio Theory (MPT)

- Harry Markowitz introduced diversification & portfolio selection
- Defined Risk as Variance/Std. Dev. (Total Risk)
- Efficient Frontier concept

1970s – Efficient Market Hypothesis (EMH) & CAPM

- Sharpe, Lintner, Mossin built on MPT
- Eugene Fama: Efficient Market Hypothesis
- Risk split: Systematic vs. Unsystematic
- Introduced Beta (β)
- CAPM: $E(Ri) = Rf + \beta i\ (E(Rm) - Rf)$

1980s – Arbitrage Pricing Theory (APT)

- Stephen Ross: Multi-factor model
- Prices reflect multiple risk factors (inflation, GDP, interest rates)
- More flexible than CAPM

1990s – Value at Risk (VaR)

- Risk management tool
- Measures maximum potential loss at confidence level
- Widely used by banks & regulators

2000s & Beyond – Post-Crisis

- Focus on systemic risk, liquidity risk, counterparty risk
- Behavioral Finance (human biases)
- New tools: Stress testing, CoVaR, Expected Shortfall
- Emphasis on interconnectedness

RANDOM WALK OF ASSET RETURNS

Future Price Movements Are Unprediitable



Random Walk and Financial Returns

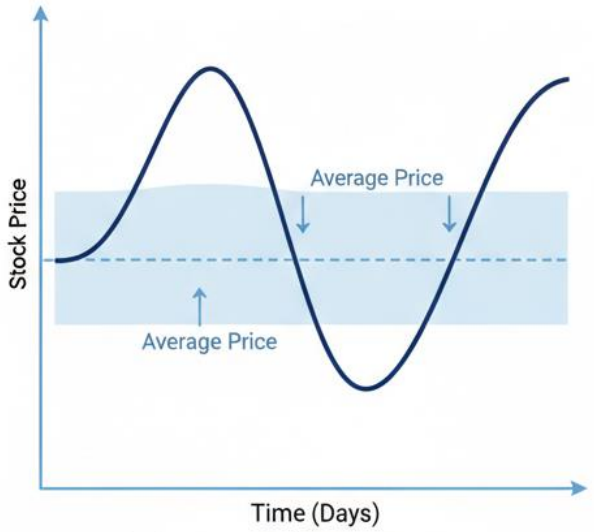
Asset prices follow a **random walk -** meaning that their future path cannot be predicted based on their past movements. Why?

Because new information, which drives stock prices, arrives randomly and unpredictably. If prices were predictable, everyone would exploit that pattern, and the pattern would quickly disappear.

Random Walk and Financial Returns

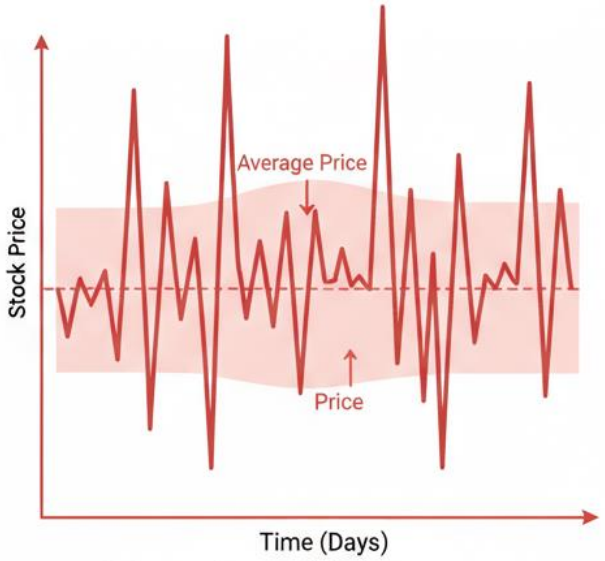
- **Returns:** We don't analyze a stock's price path directly, but rather its returns (the percentage change in price). Financial theory, specifically the Efficient Market Hypothesis, suggests that these returns are independent and identically distributed (i.i.d.).
- **Mean:** The "drift" in a random walk is the **expected return** or **mean return** of the asset. Over the long run, we expect stocks to have a positive mean return to compensate investors for risk.
- **Variance/Standard Deviation:** The "random" part of the walk is the volatility or risk, which we measure with variance or its square root, standard deviation. A larger variance means the random steps are bigger, leading to wider fluctuations in returns.

Low Standard Deviation: Stable Stock



Price stays close to average
= Low Voltality

High Standard Deviation: Volatile Stock

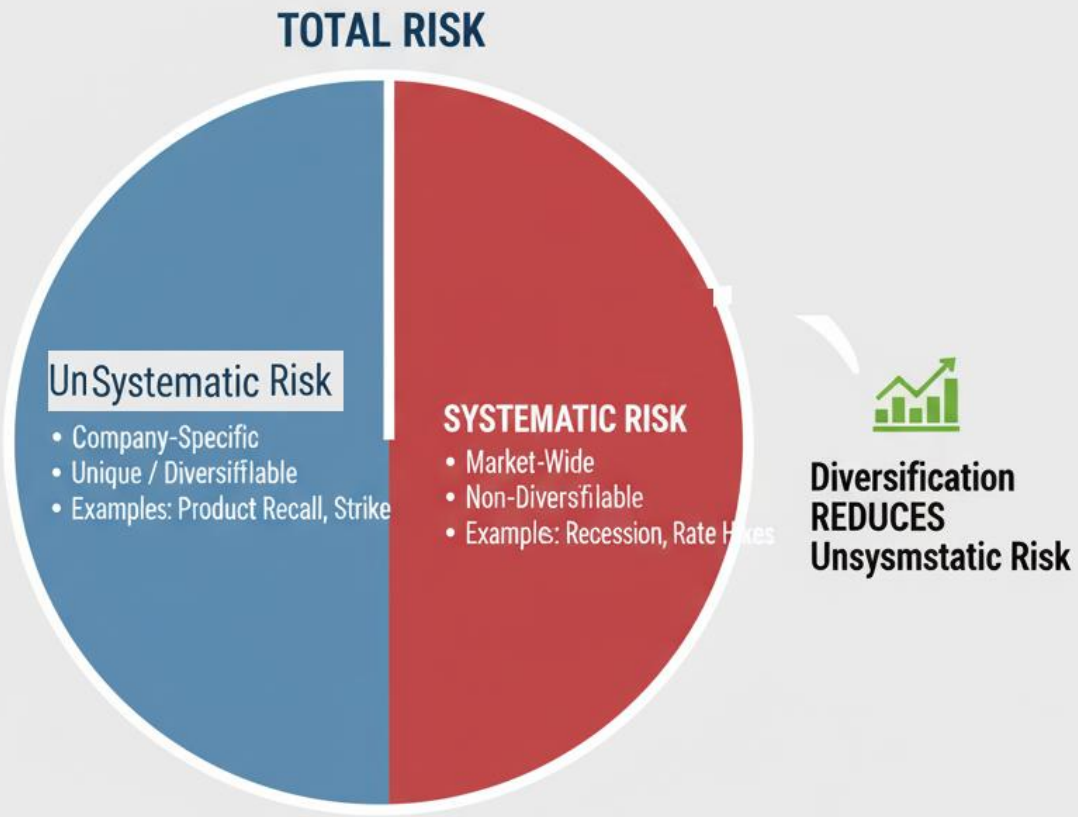


Price swings widely from average
average = High Voltality

Standard Deviation = Total Risk (Market + Company Factors)

VISUALIZING TOTAL RISK DECOMPOSITION

Total Risk = Systematic Risk + UnSystematic Risk



Systematic Risk is measured by BETA

Visualizing Beta: Market Senstiivity

Systematic Risk



Beta = Market Risk (Non-Diversivable Risk)

Measuring Risk

Variance – Average (expected) value of squared deviations from mean. A measure of volatility.

Standard Deviation – Square root of variance. A measure of volatility.

Measuring Risk

Coin Toss Game-calculating variance and standard deviation

(1) Percent Rate of Return	(2) Deviation from Mean	(3) Squared Deviation
+40	+30	900
+10	0	0
+10	0	0
-20	-30	900

Variance=average of squared deviations=1800/4=450

Standard deviation=square of root variance= $\sqrt{450}$=21.2%

Measuring Risk

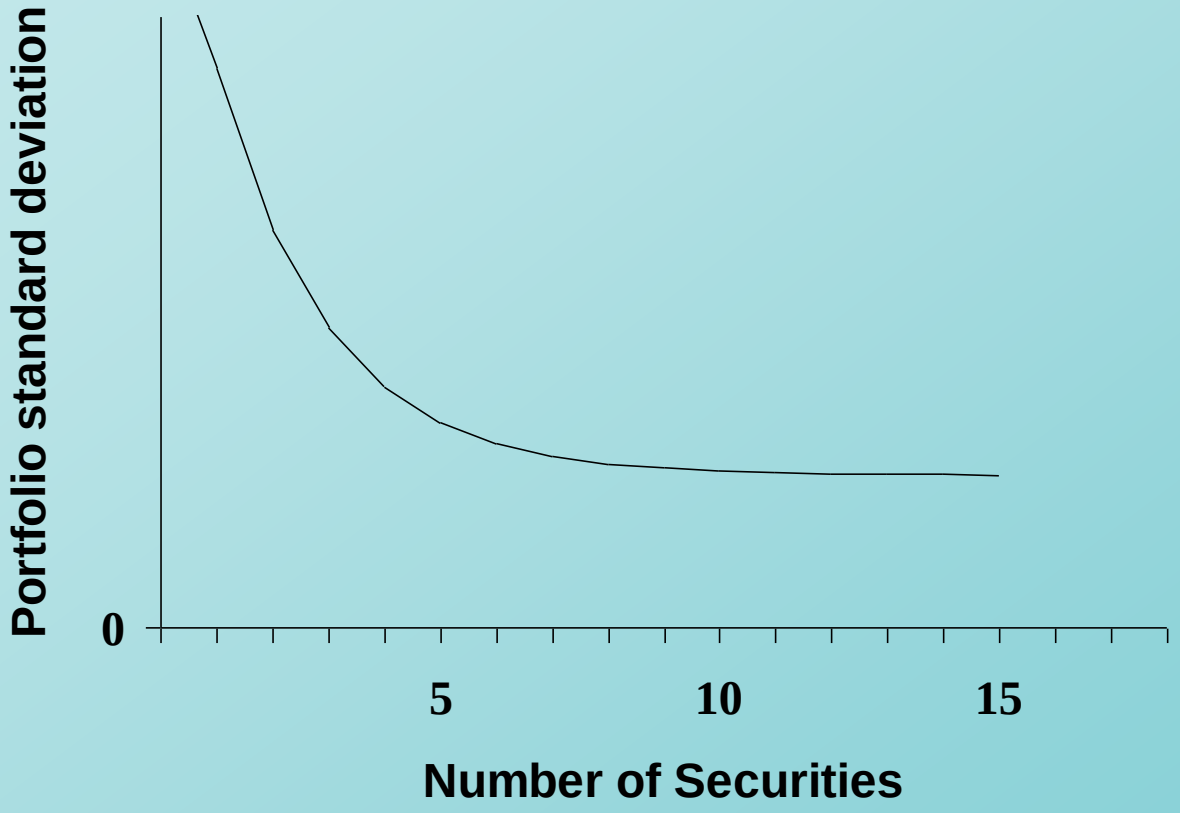
Diversification - Strategy designed to reduce risk by spreading the portfolio across many investments.

Unique Risk - Risk factors affecting only that firm. Also called "diversifiable risk."

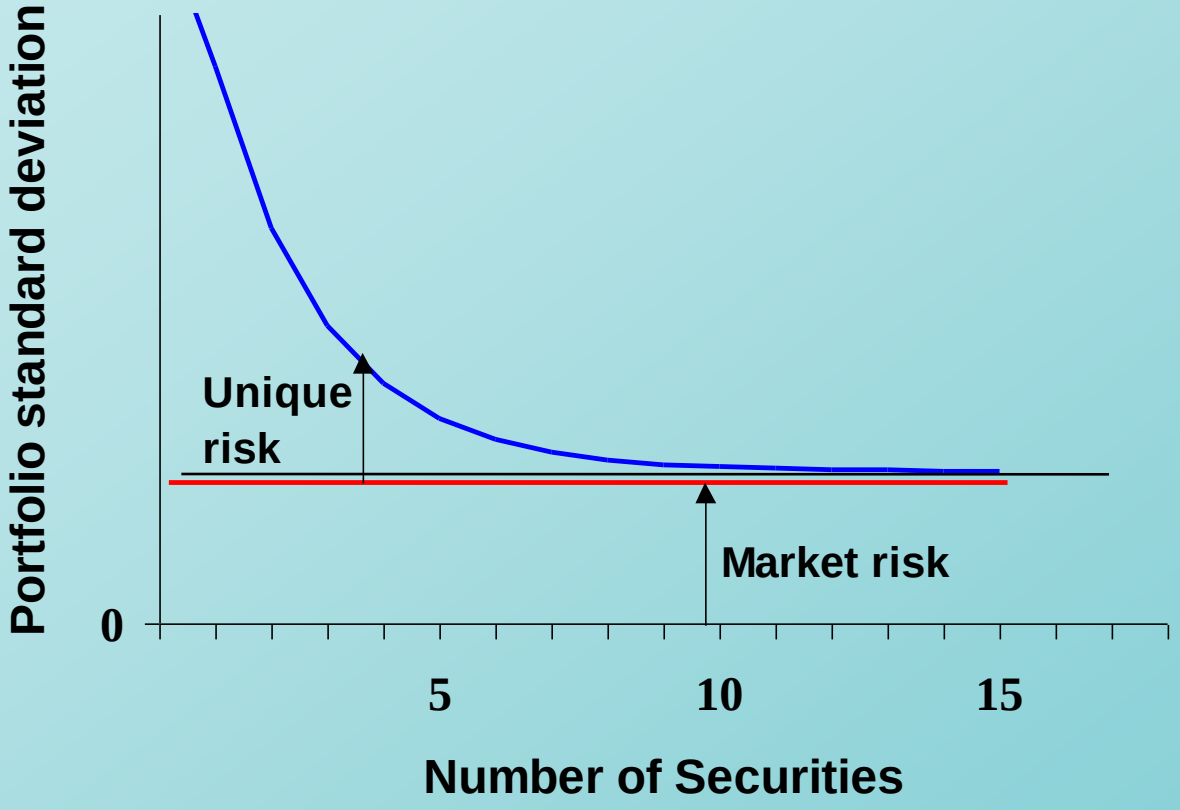
Market Risk - Economy-wide sources of risk that affect the overall stock market. Also called "systematic risk."

- Systematic Risks (non-diversifiable risk; market risk; relevant risk) :
  - Interest rate risk
  - Inflation risk
  - Maturity risk
  - Liquidity risk
  - Exchange rate risk
  - Political risk
- Unsystematic Risk (diversifiable risk, firm specific risk)
  - Business Risk
  - Financial risk
  - Default risk
- Combined risk (some systematic risk and some unsystematic risk)
  - Total risk (Stand-alone risk)
  - Corporate risk

Measuring Risk



Measuring Risk



Portfolio rate of return		fraction of portfolio in first asset	*x*	rate of return on first asset
	+	fraction of portfolio in second asset	*x*	rate of return on second asset

Portfolio Risk

The variance of a two stock portfolio is the sum of these four boxes:

	Stock 1	Stock 2
Stock 1	$x_1^2 \sigma_1^2$	$x_1 x_2 \sigma_{12} = x_1 x_2 \rho_{12} \sigma_1 \sigma_2$
Stock 2	$x_1 x_2 \sigma_{12} = x_1 x_2 \rho_{12} \sigma_1 \sigma_2$	$x_2^2 \sigma_2^2$

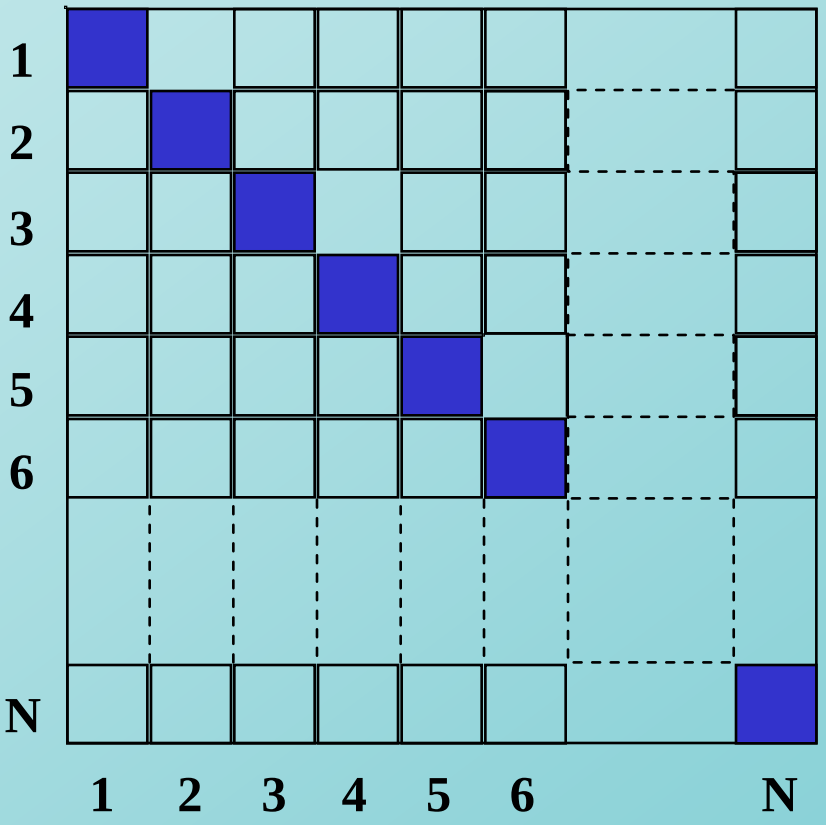
Portfolio Risk

$$\text{Expected Portfolio Return} = (x_1 r_1) + (x_2 r_2)$$

$$\text{Portfolio Variance} = x_1^2 \sigma_1^2 + x_2^2 \sigma_2^2 + 2(x_1 x_2 \rho_{12} \sigma_1 \sigma_2)$$

Portfolio Risk

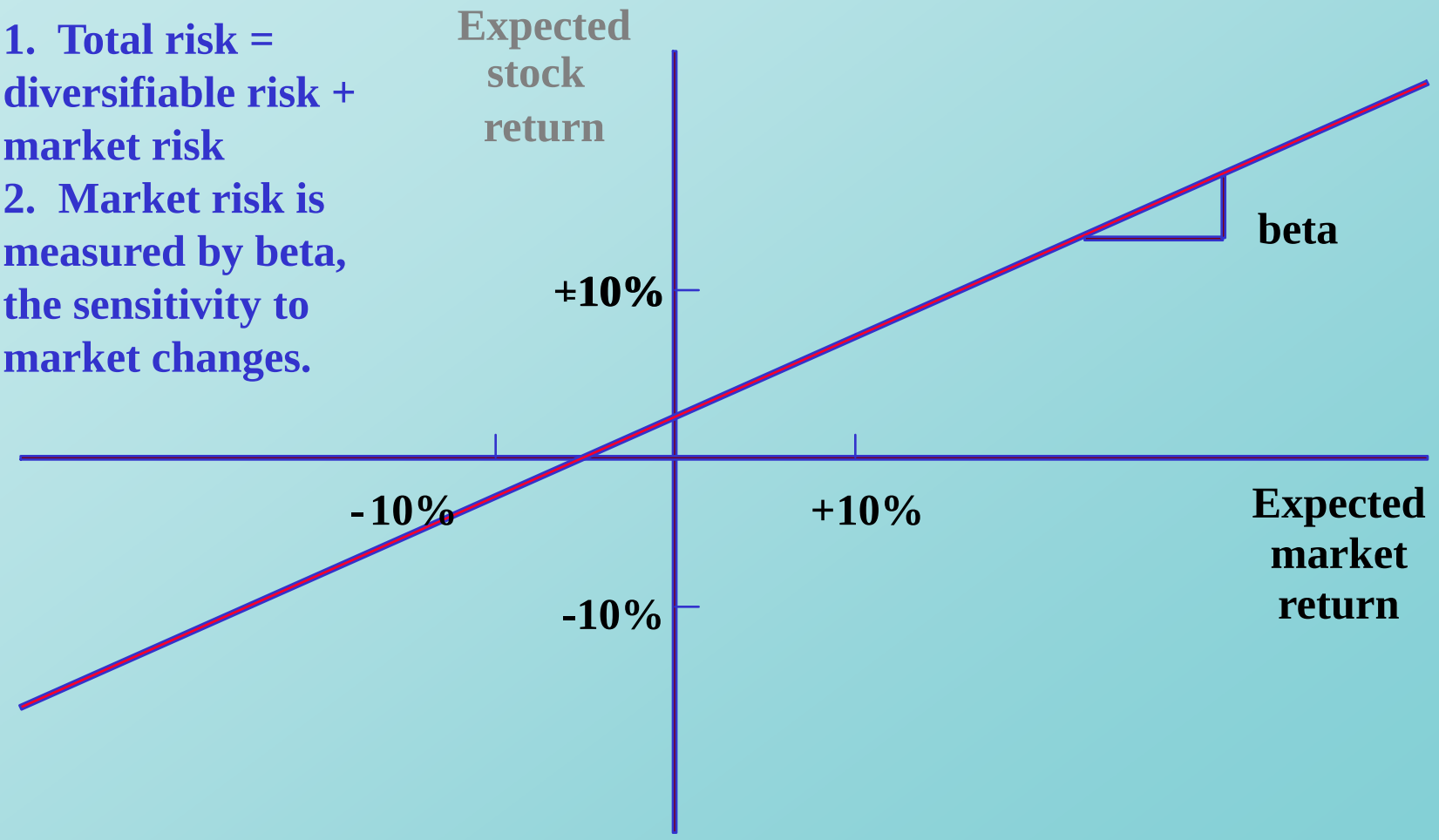
The shaded boxes contain variance terms; the remainder contain covariance terms.



To calculate portfolio variance add up the boxes

Beta and Unique Risk

1. Total risk = diversifiable risk + market risk
2. Market risk is measured by beta, the sensitivity to market changes.



Beta and Unique Risk

Market Portfolio - Portfolio of all assets in the economy. In practice a broad stock market index, such as the S&P Composite, is used to represent the market.

Beta - Sensitivity of a stock's return to the return on the market portfolio.

Formula for stock & market return

Daily Variance =var.s(Stock Return)

Annual Variance =Daily Variance x Days

Annual S.D. =sqrt(Annual Variance)

Annual Covariance =covariance.s(Stock Return, Market Return) x Days

Correlation = correl(Stock Return, Market Return)

Beta =slope(Stock Return, Market Return)

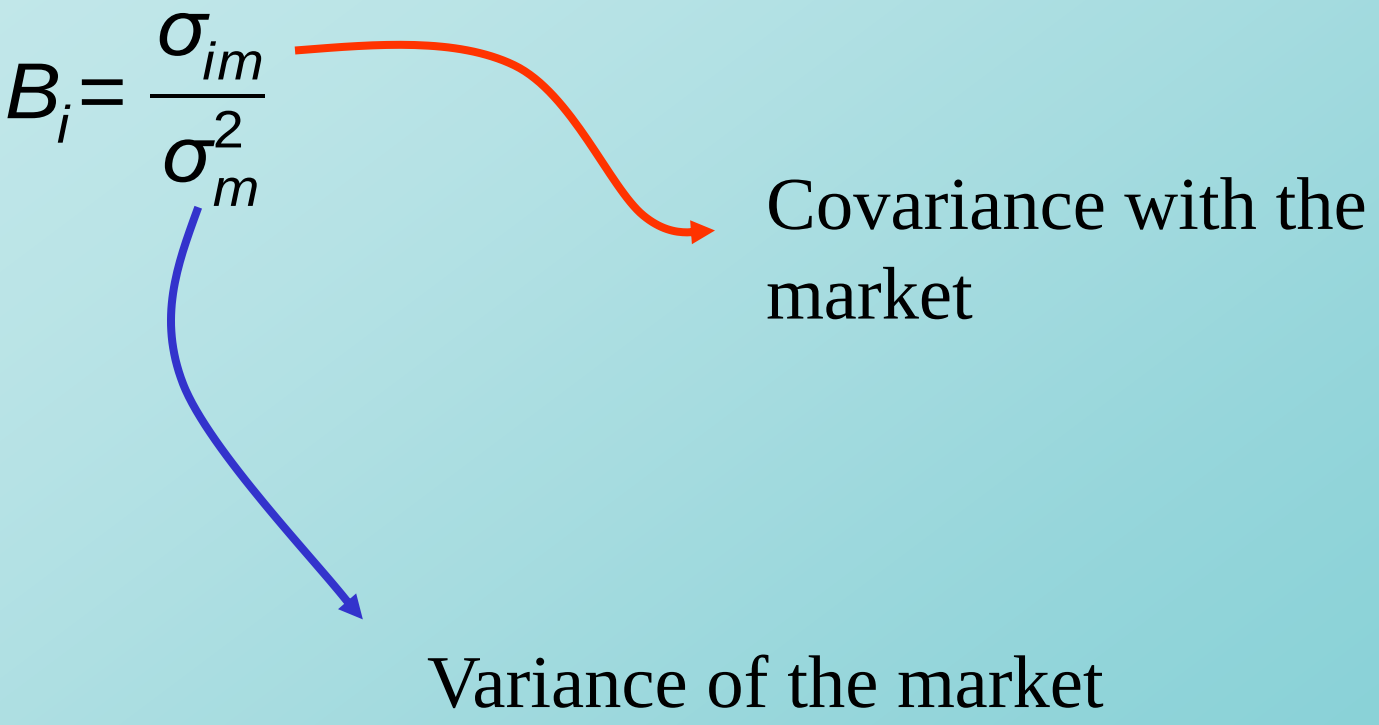
Beta =covariance.s(Stock Return,Market Return)/var.s(Market Return)

Beta =Correlation x S.D. of stock/S.D. of market

Beta and Unique Risk

$$B_i = \frac{\sigma_{im}}{\sigma_m^2}$$

Beta and Unique Risk



Markowitz Portfolio Theory

Combining stocks into portfolios can reduce standard deviation below the level obtained from a simple weighted average calculation.

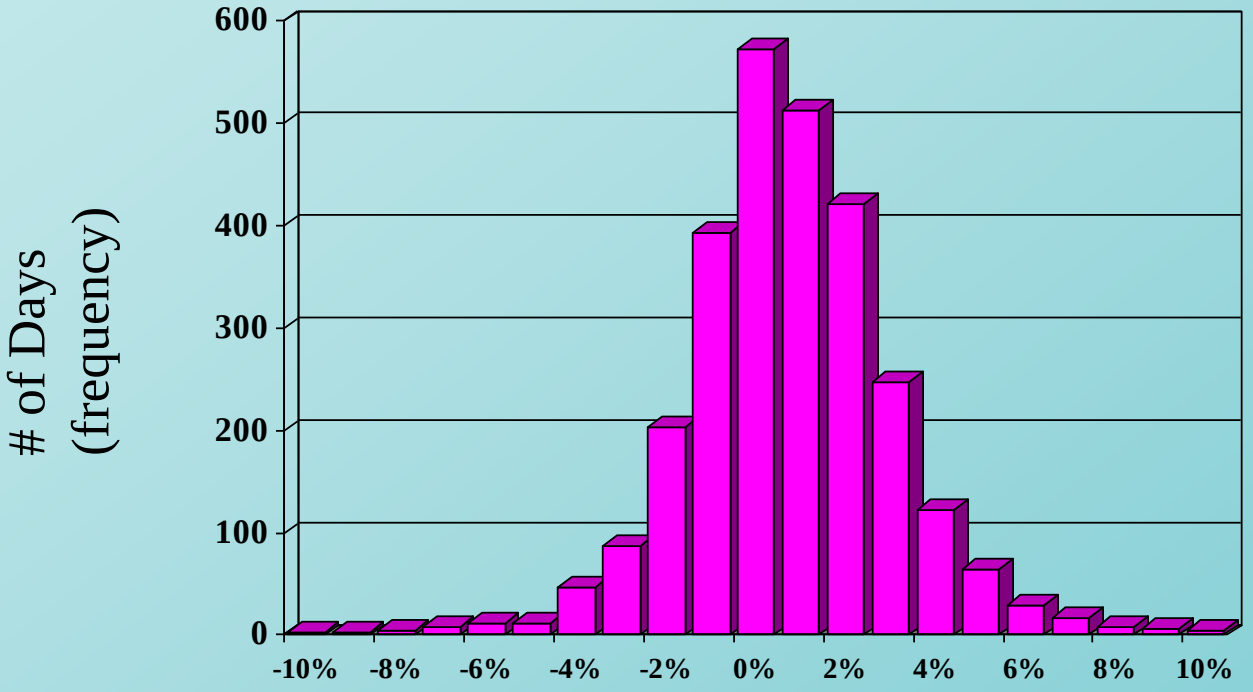
Correlation coefficients make this possible.

The various weighted combinations of stocks that create this standard deviations constitute the set of ***efficient portfolios***.

Markowitz Portfolio Theory

Price changes vs. Normal distribution

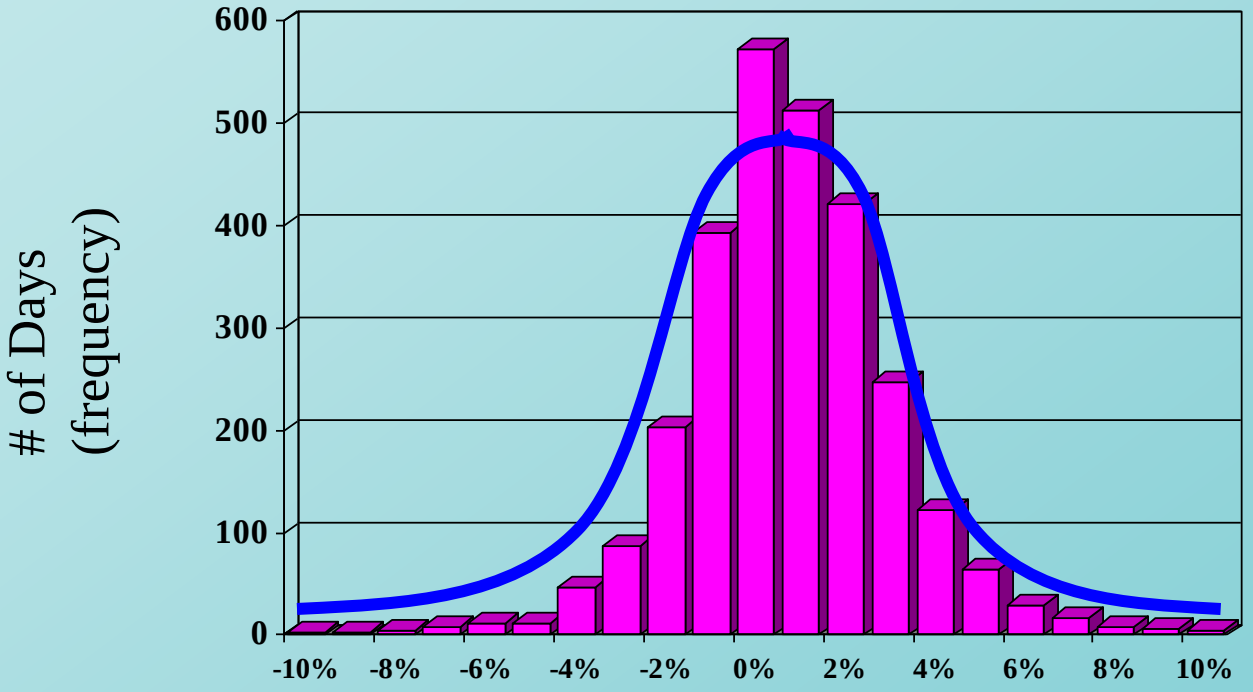
Microsoft - Daily % change 1986-1997



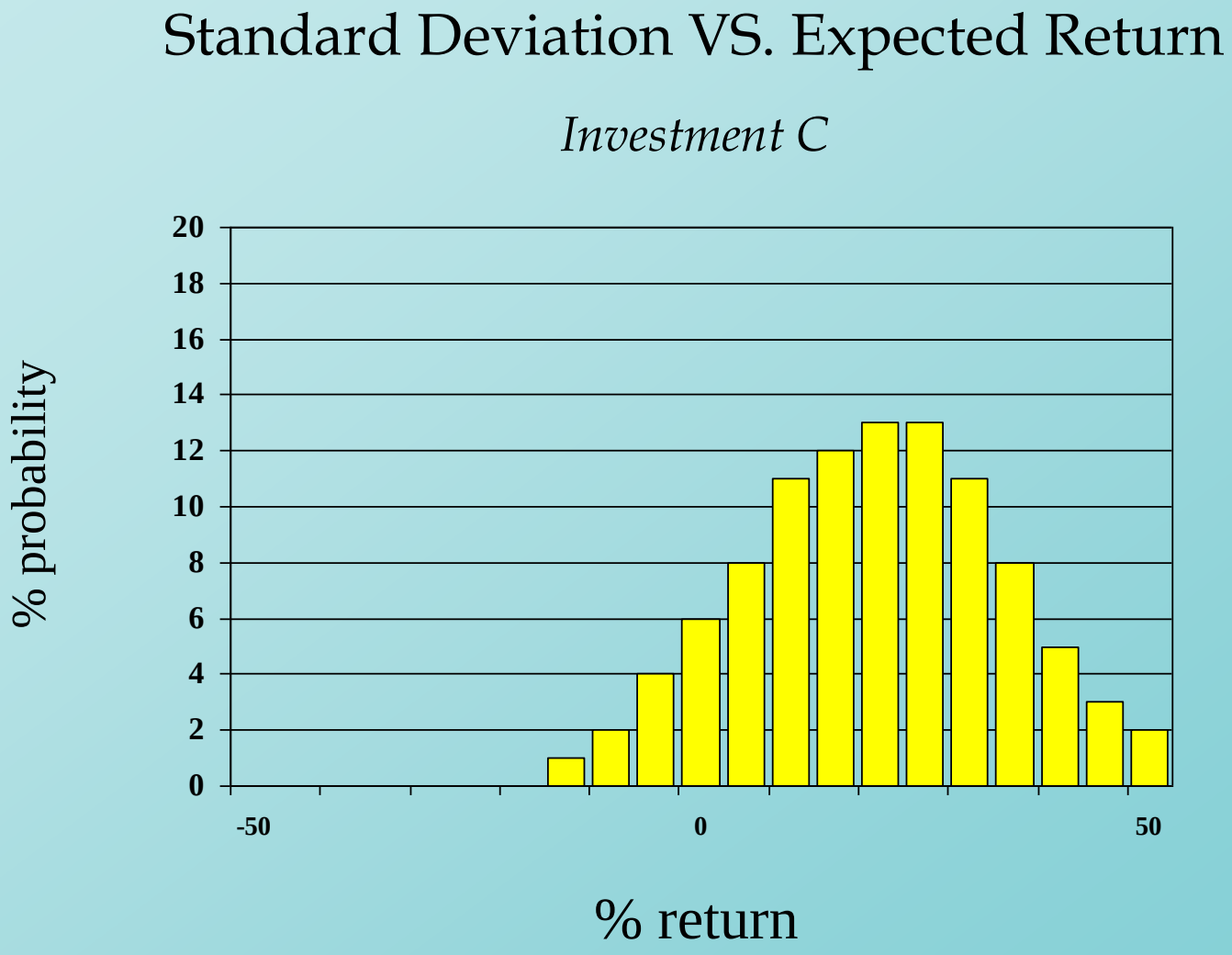
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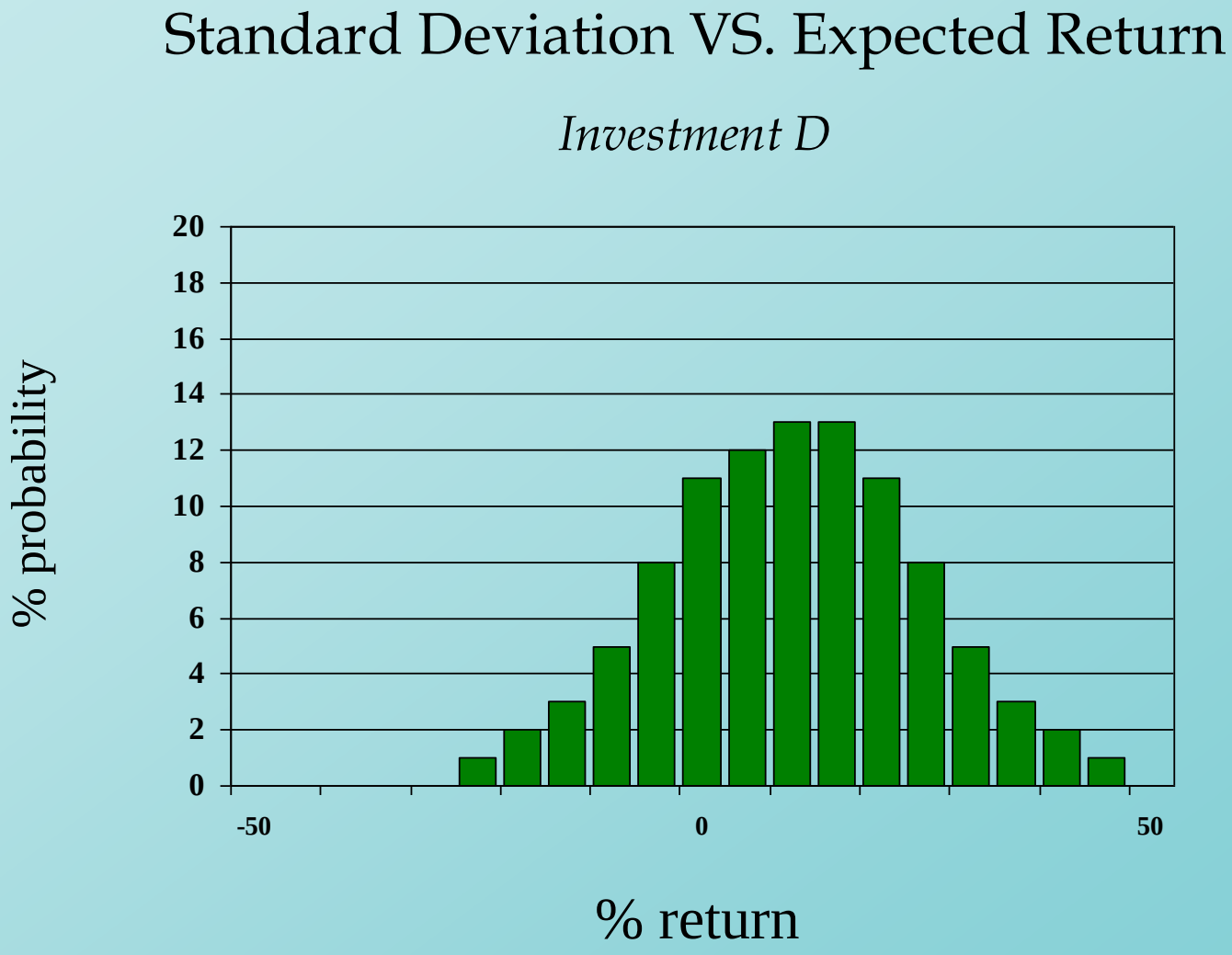
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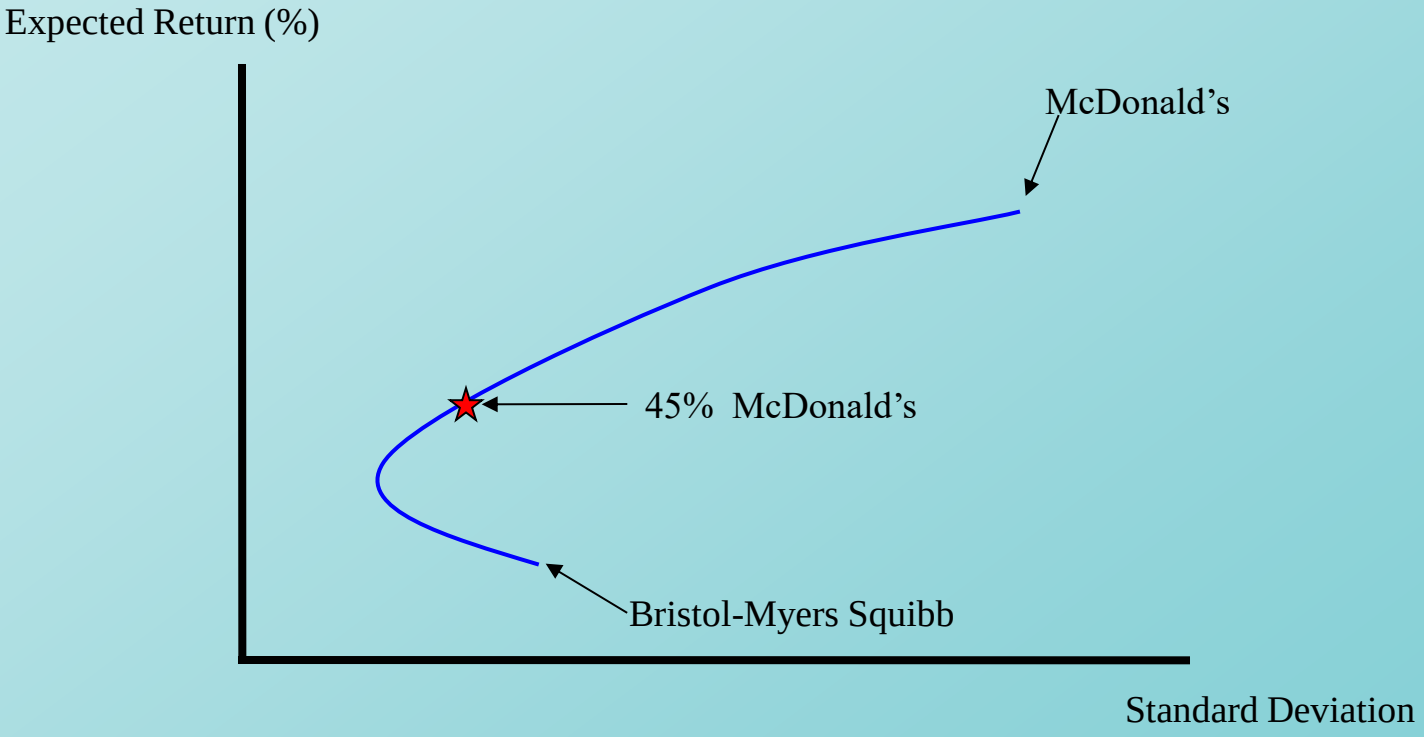


Markowitz Portfolio Theory



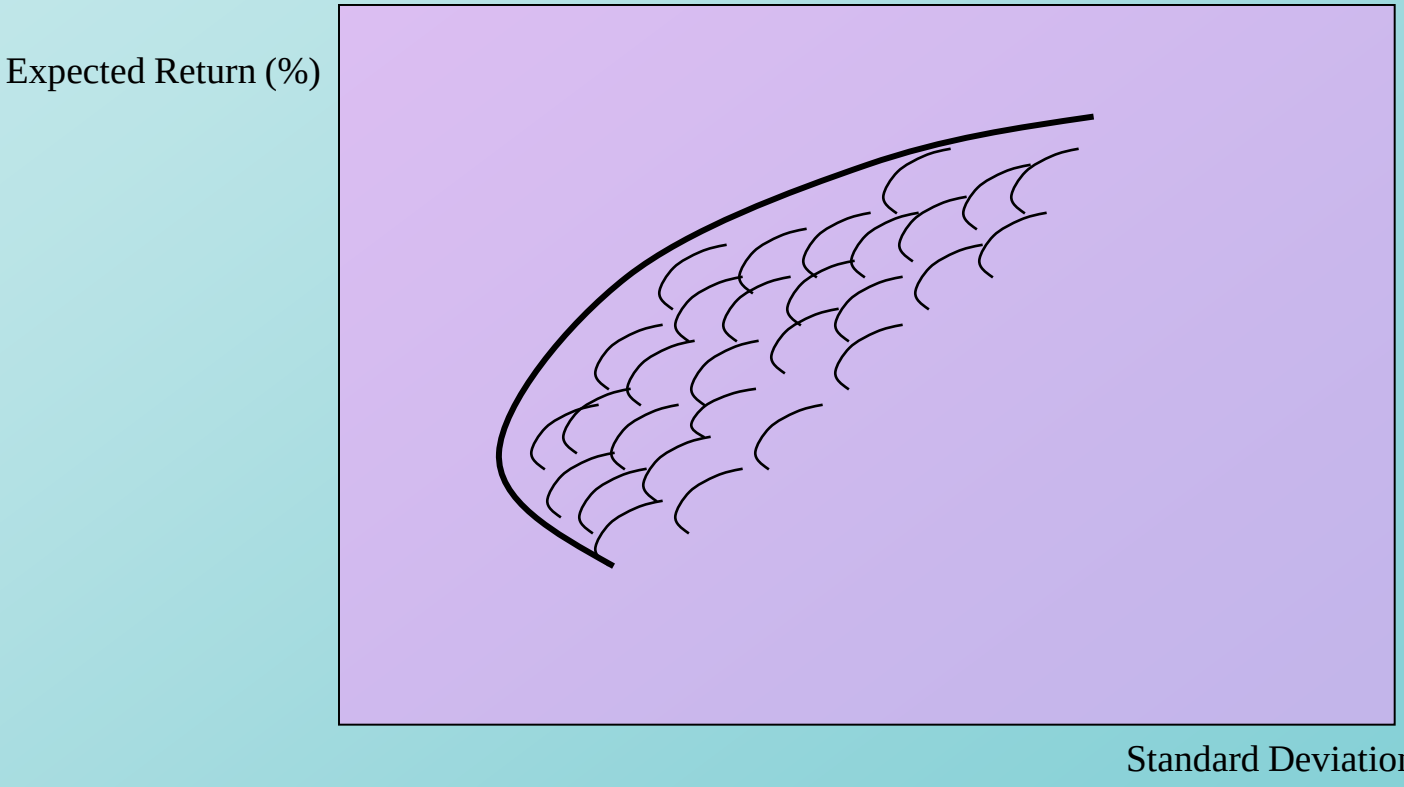
Markowitz Portfolio Theory

- Expected Returns and Standard Deviations vary given different weighted combinations of the stocks.



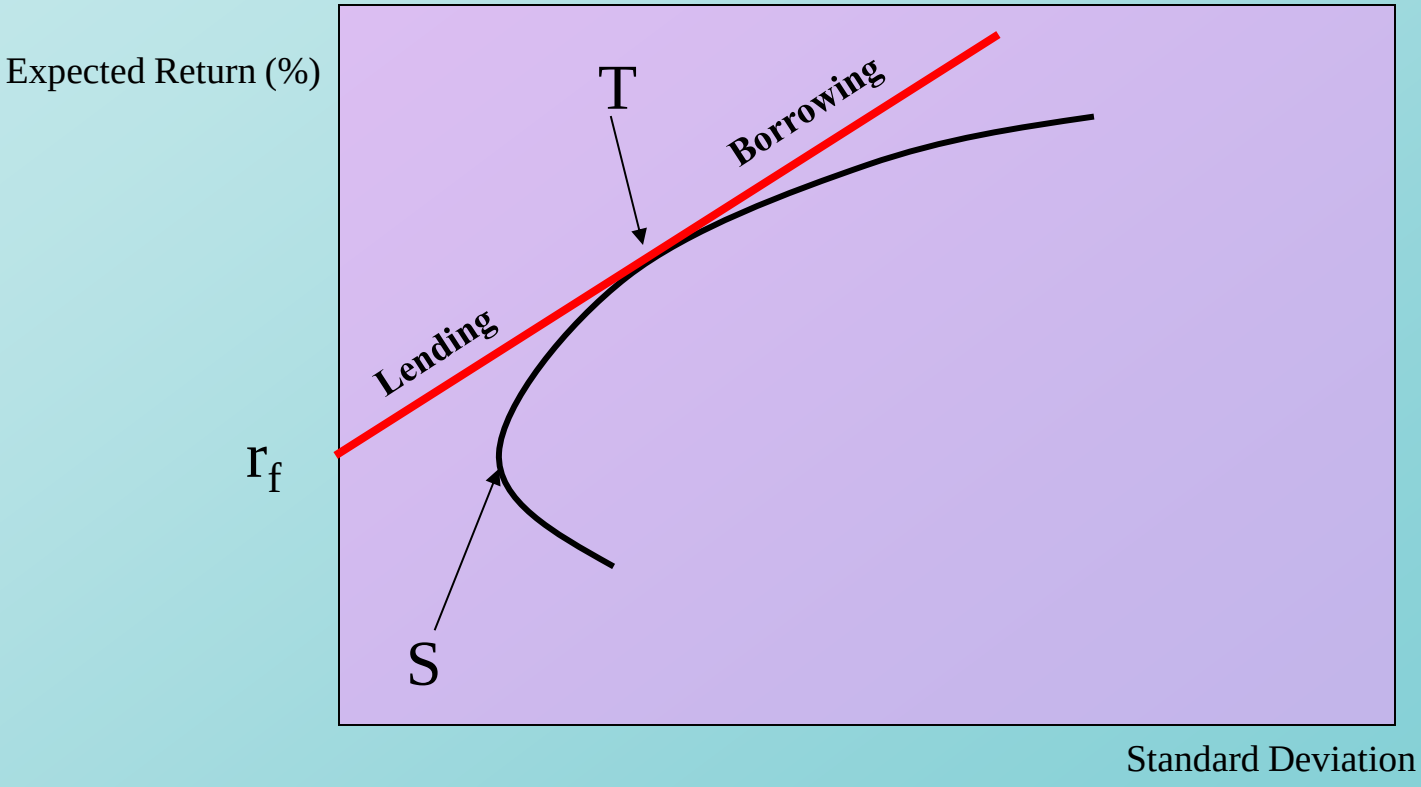
Efficient Frontier

- Each half egg shell represents the possible weighted combinations for two stocks.
- The composite of all stock sets constitutes the efficient frontier.



Efficient Frontier

•Lending or Borrowing at the risk free rate (r_f) allows us to exist outside the efficient frontier.



Efficient Frontier

Example — Correlation Coefficient = .4

Stocks	σ	% of Portfolio	Avg Return
ABC Corp	28	60%	15%
Big Corp	42	40%	21%

Standard Deviation = weighted avg = 33.6

Standard Deviation = Portfolio = 28.1

Return = weighted avg = Portfolio = 17.4%

Efficient Frontier

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Let's Add stock New Corp to the portfolio

Efficient Frontier

Example — Correlation Coefficient = .3

Stocks	σ	% of Portfolio	Avg Return
Portfolio	28.1	50%	17.4%
New Corp	***30***	***50%***	***19%***

NEW Standard Deviation = weighted avg = 31.80

NEW Standard Deviation = Portfolio = 23.43

NEW Return = weighted avg = Portfolio = 18.20%

Efficient Frontier

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NOTE: Higher return & Lower risk

Efficient Frontier

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How did we do that?

Efficient Frontier

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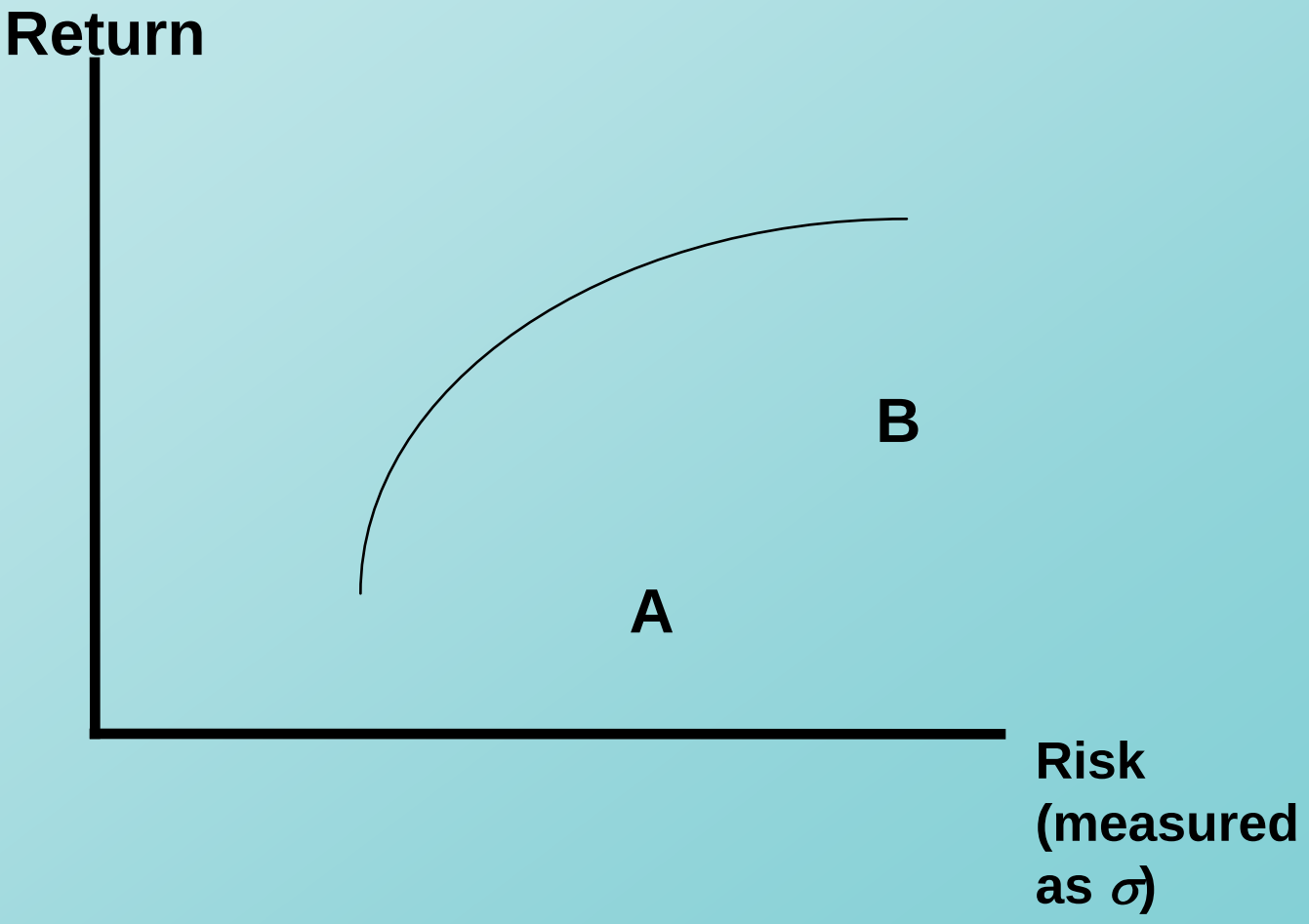
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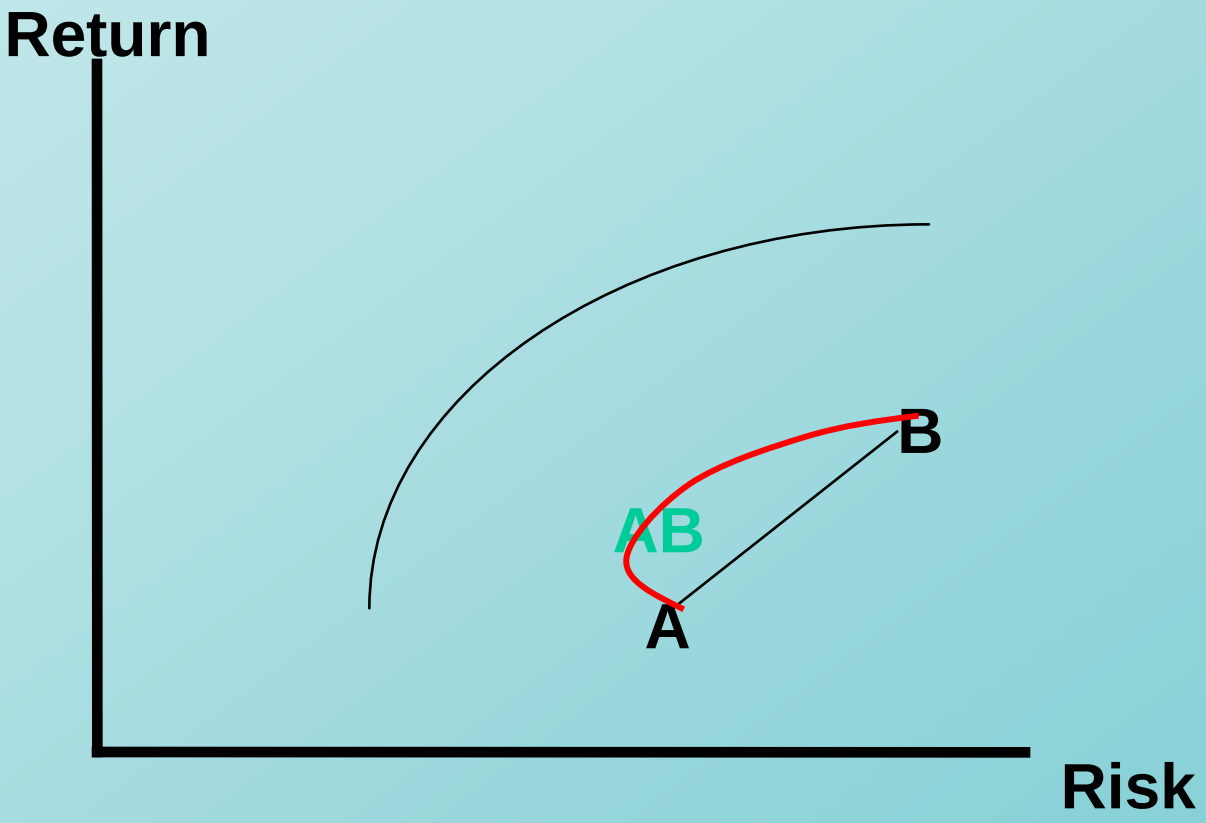
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How did we do that? DIVERSIFICATION

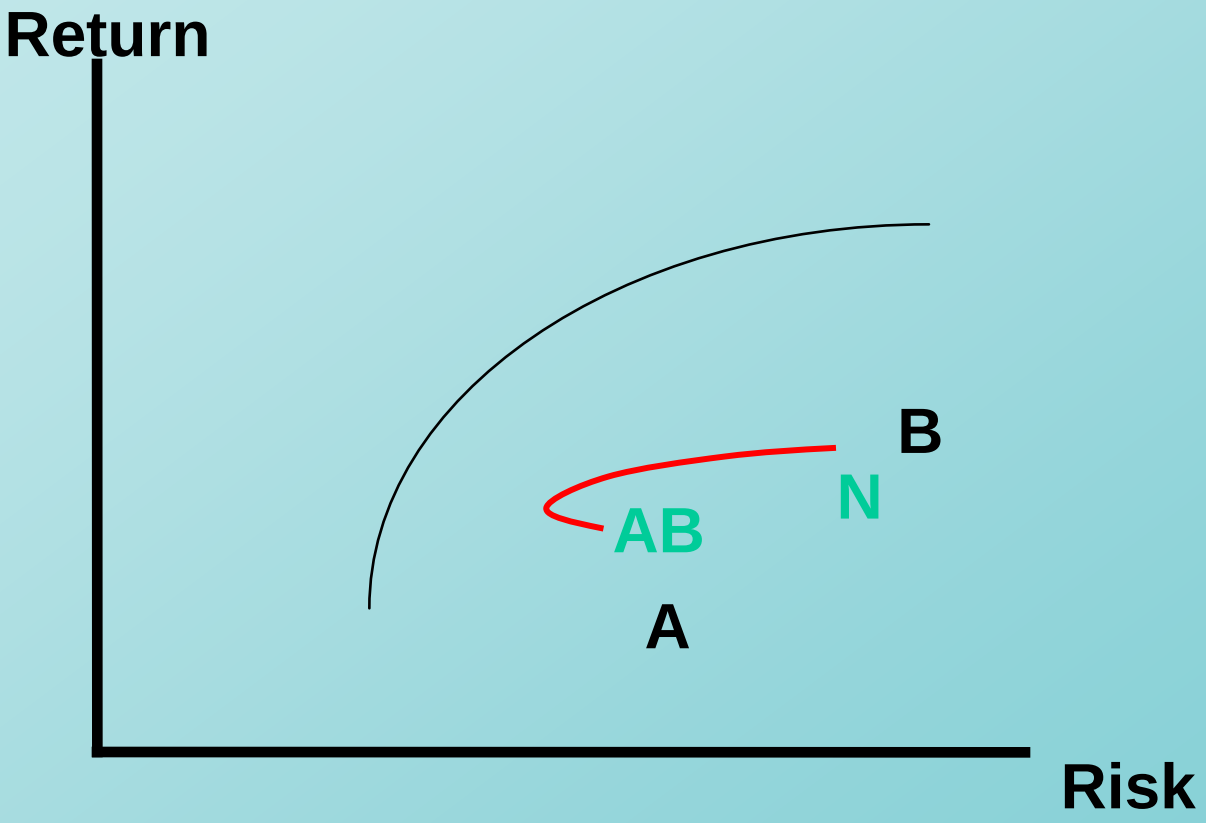
Efficient Frontier



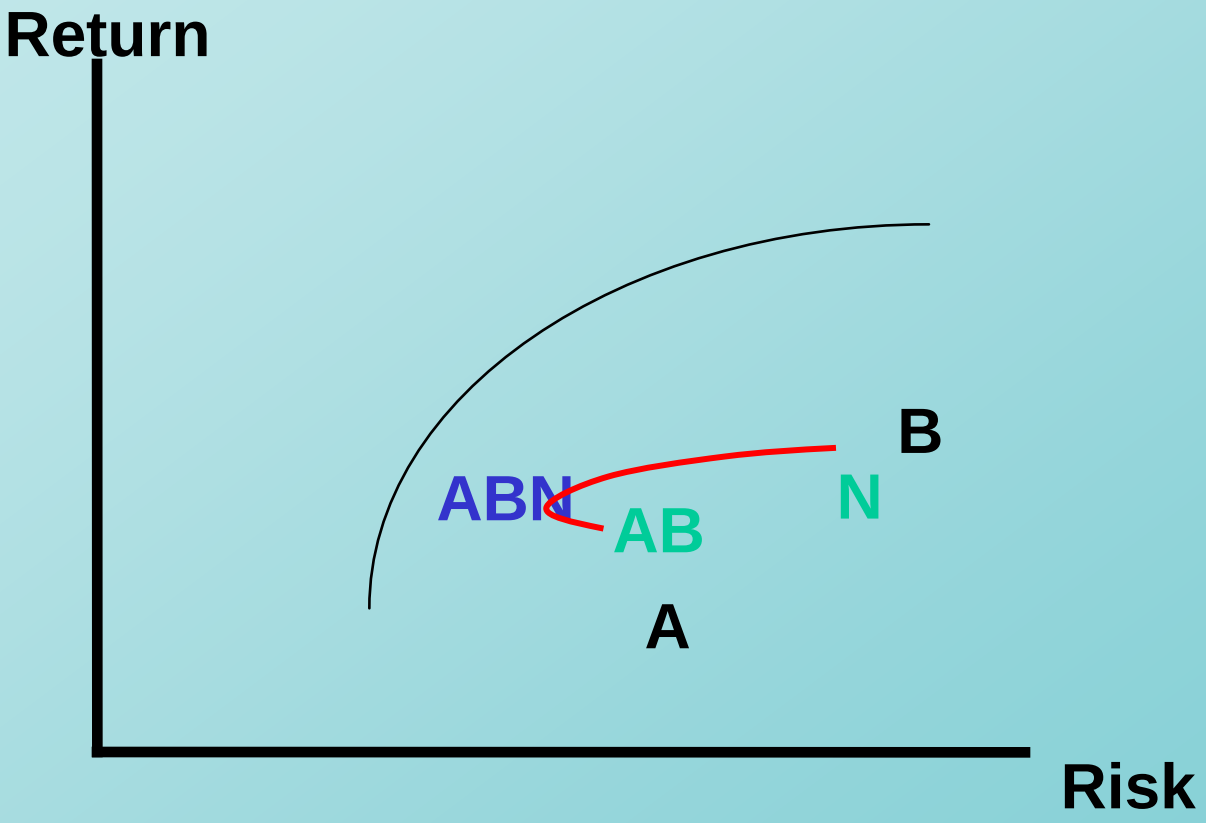
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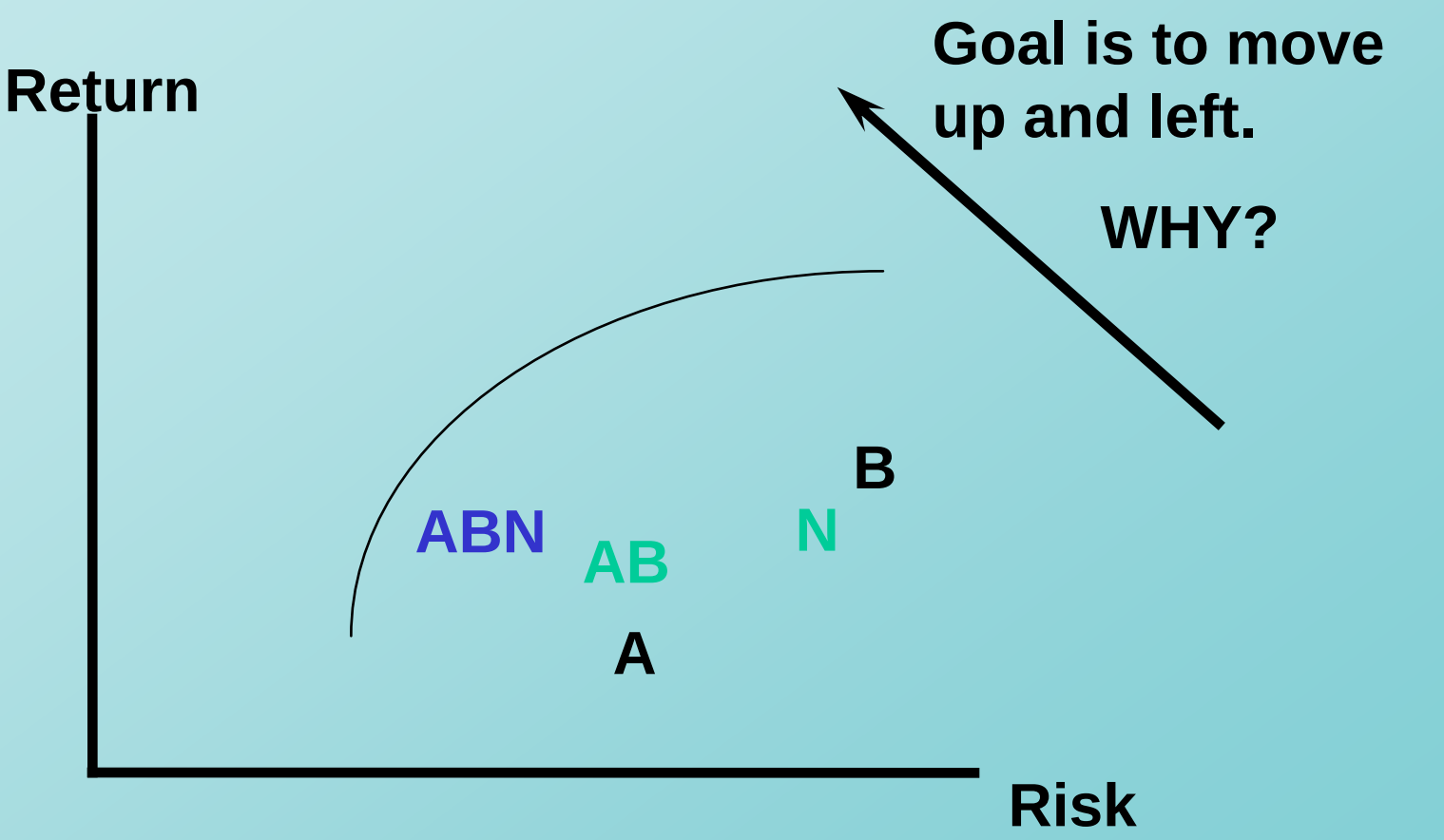
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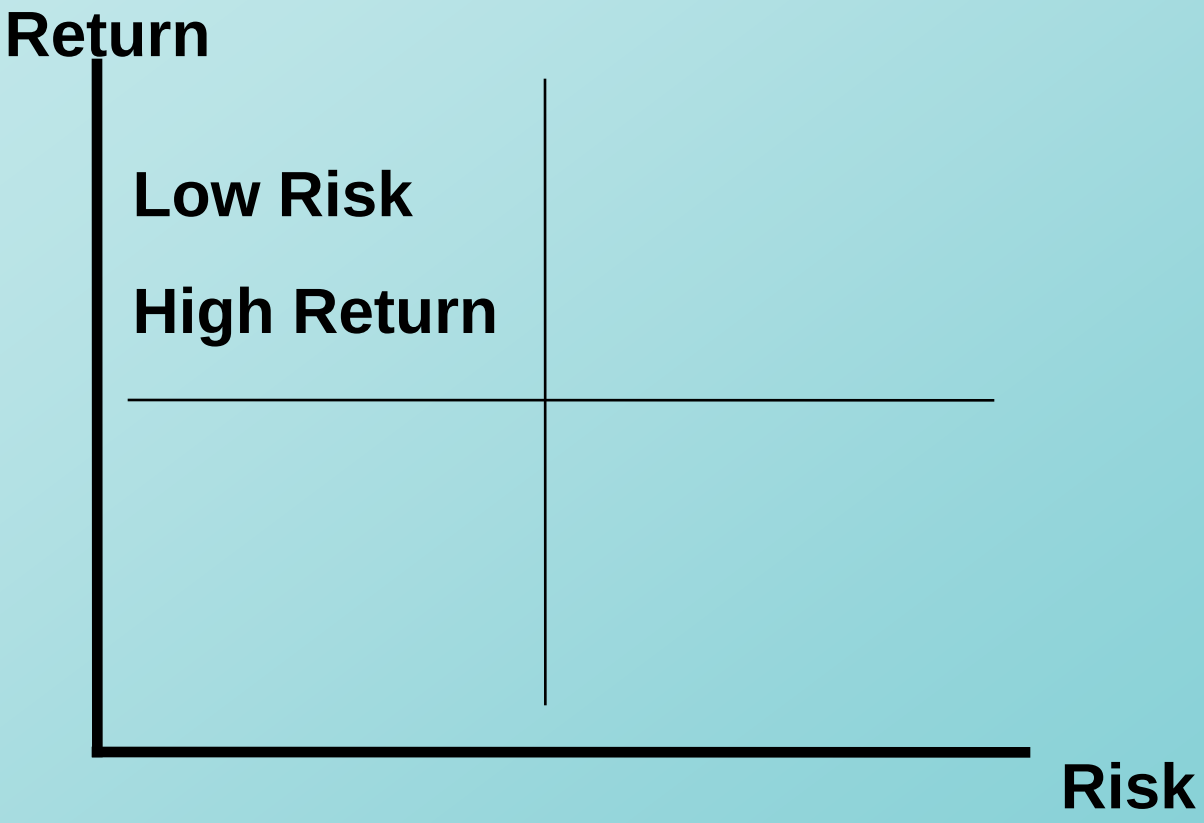
Efficient Frontier



Efficient Frontier



Efficient Frontier



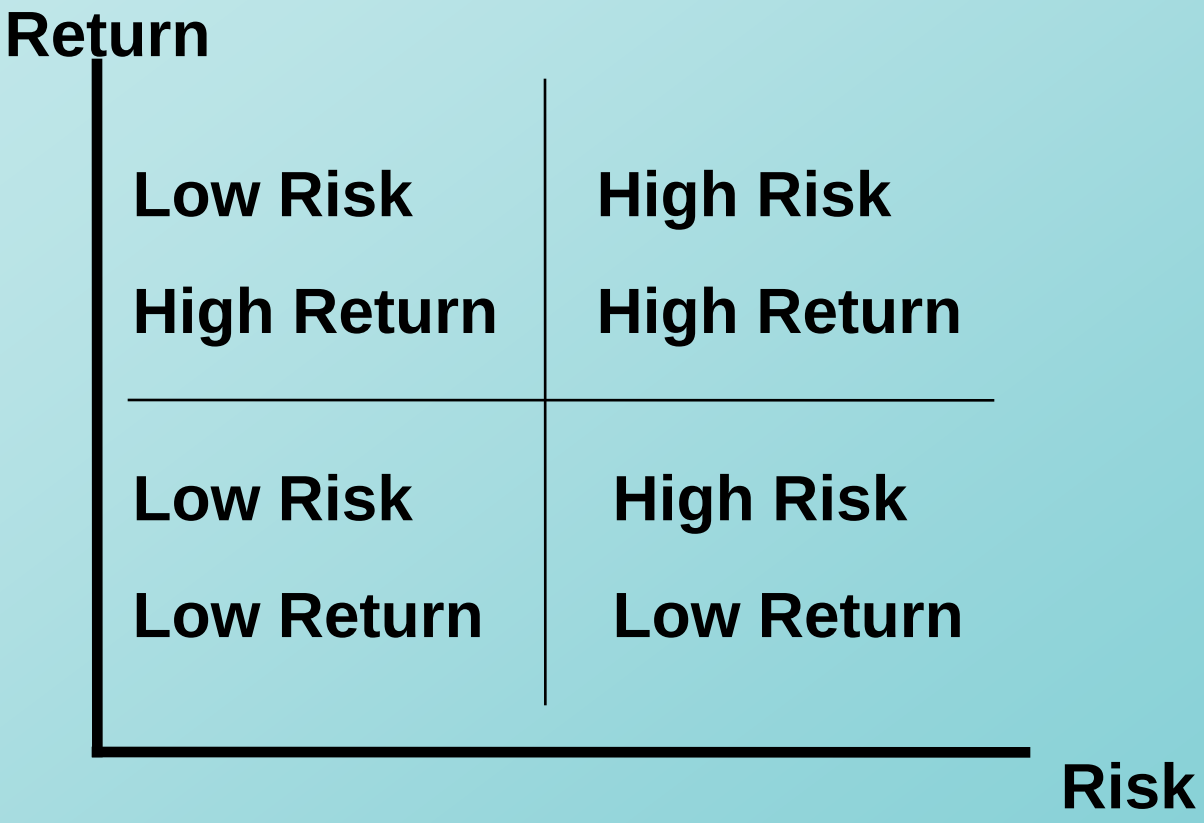
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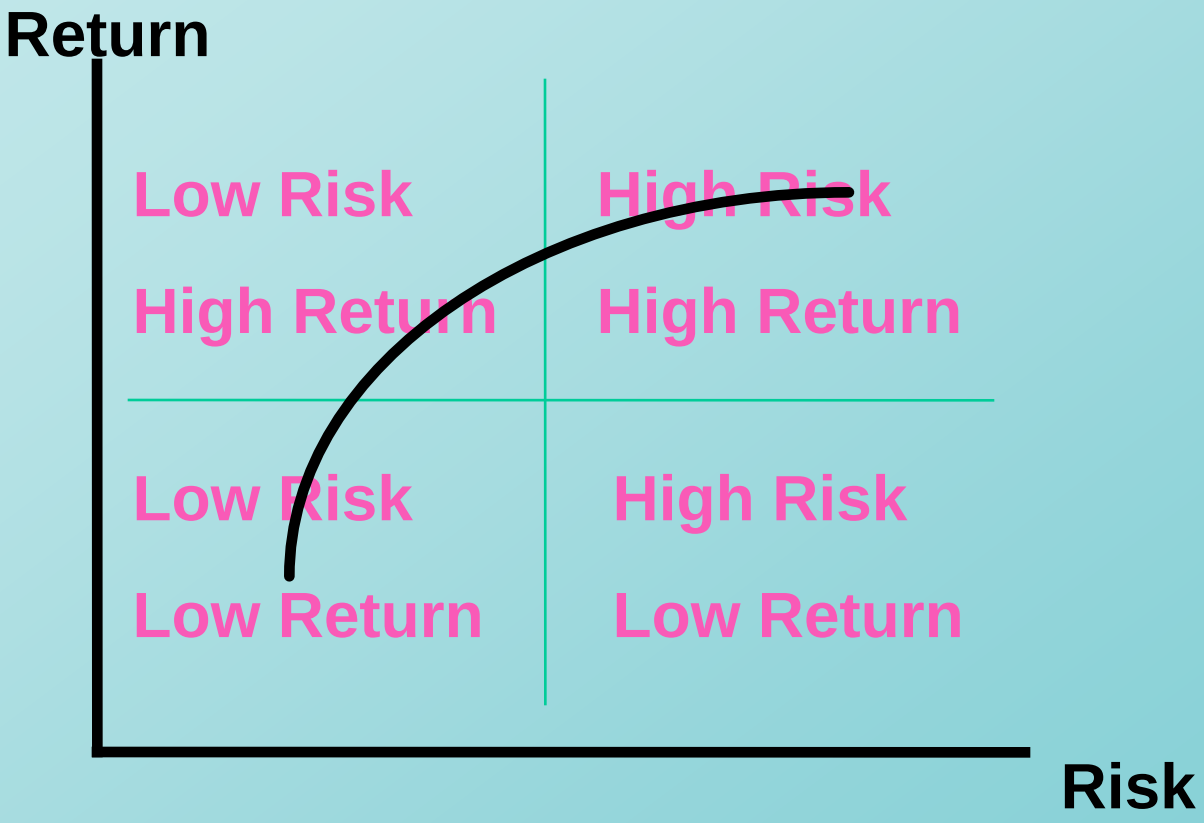
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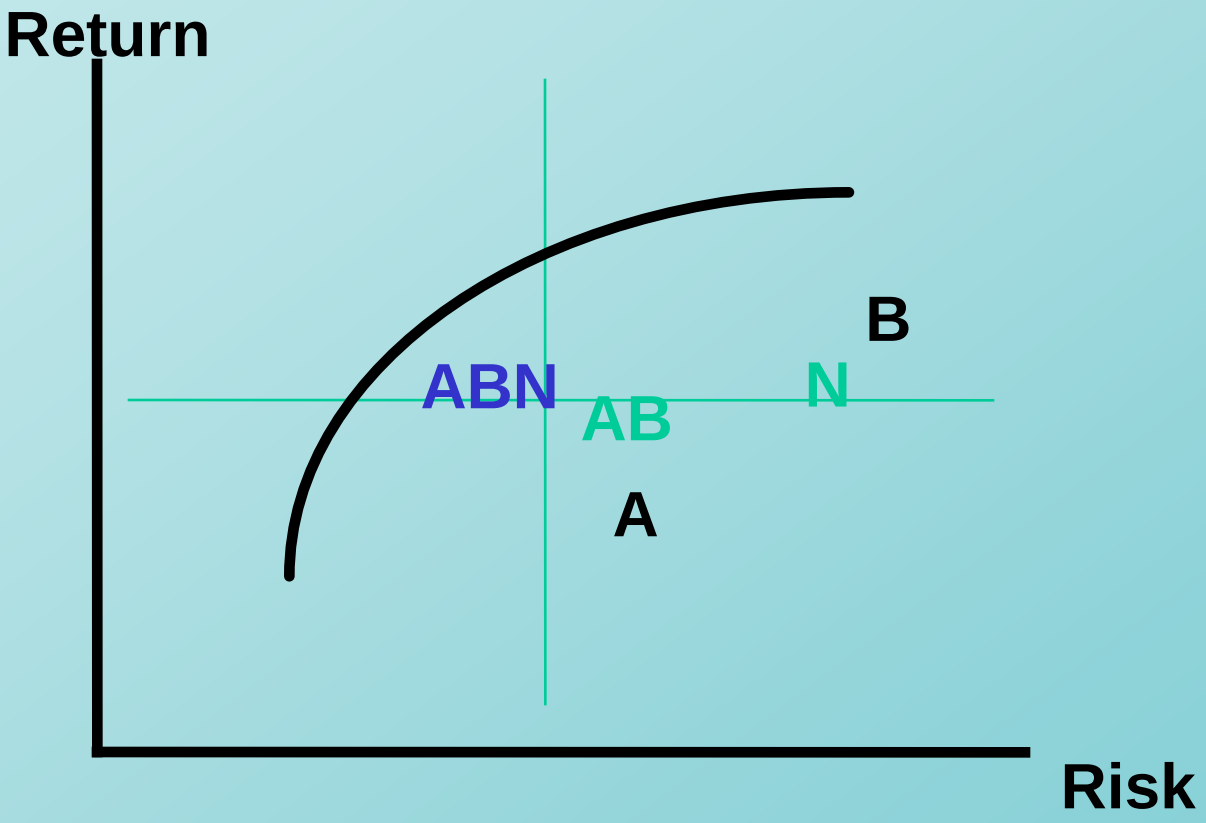
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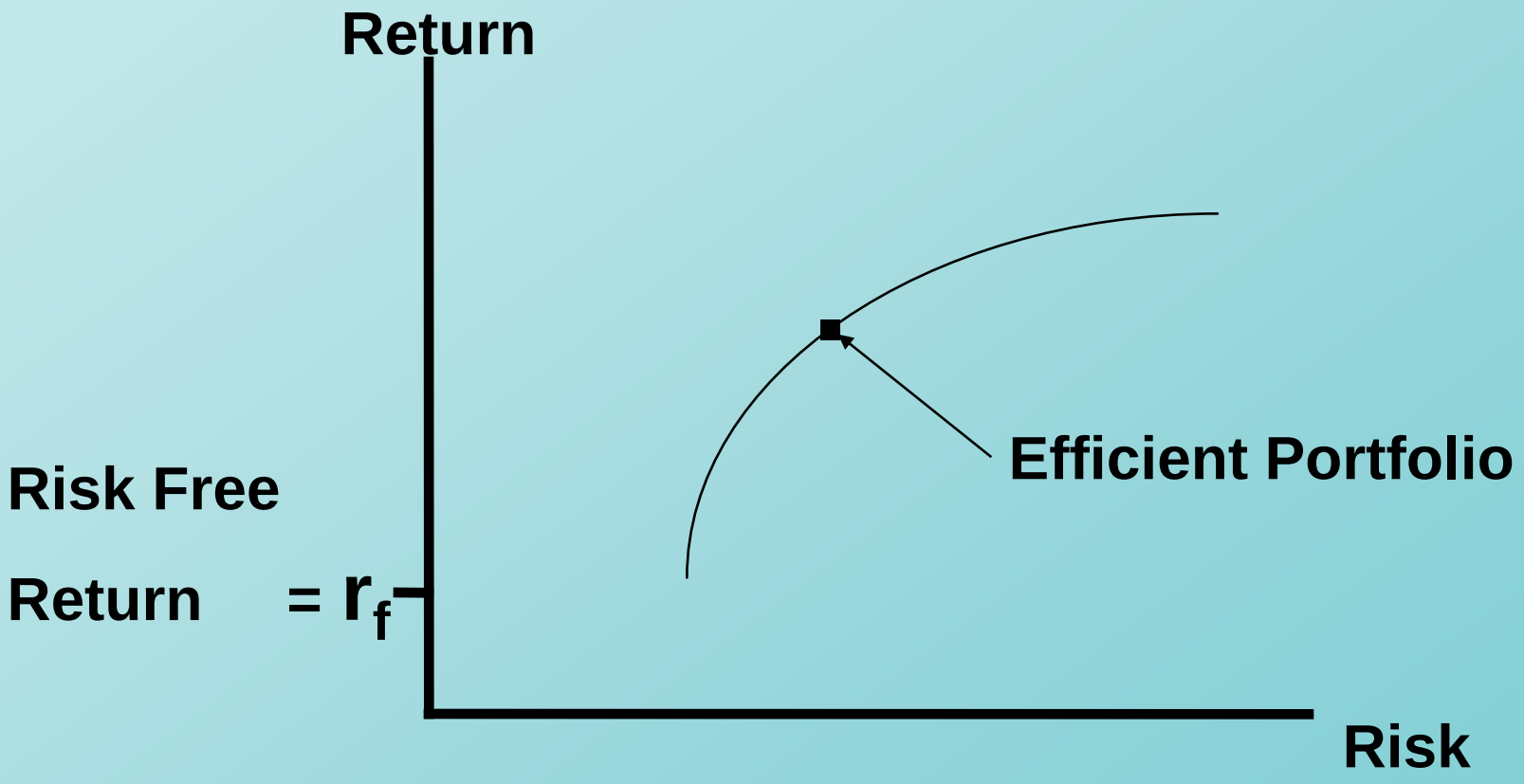
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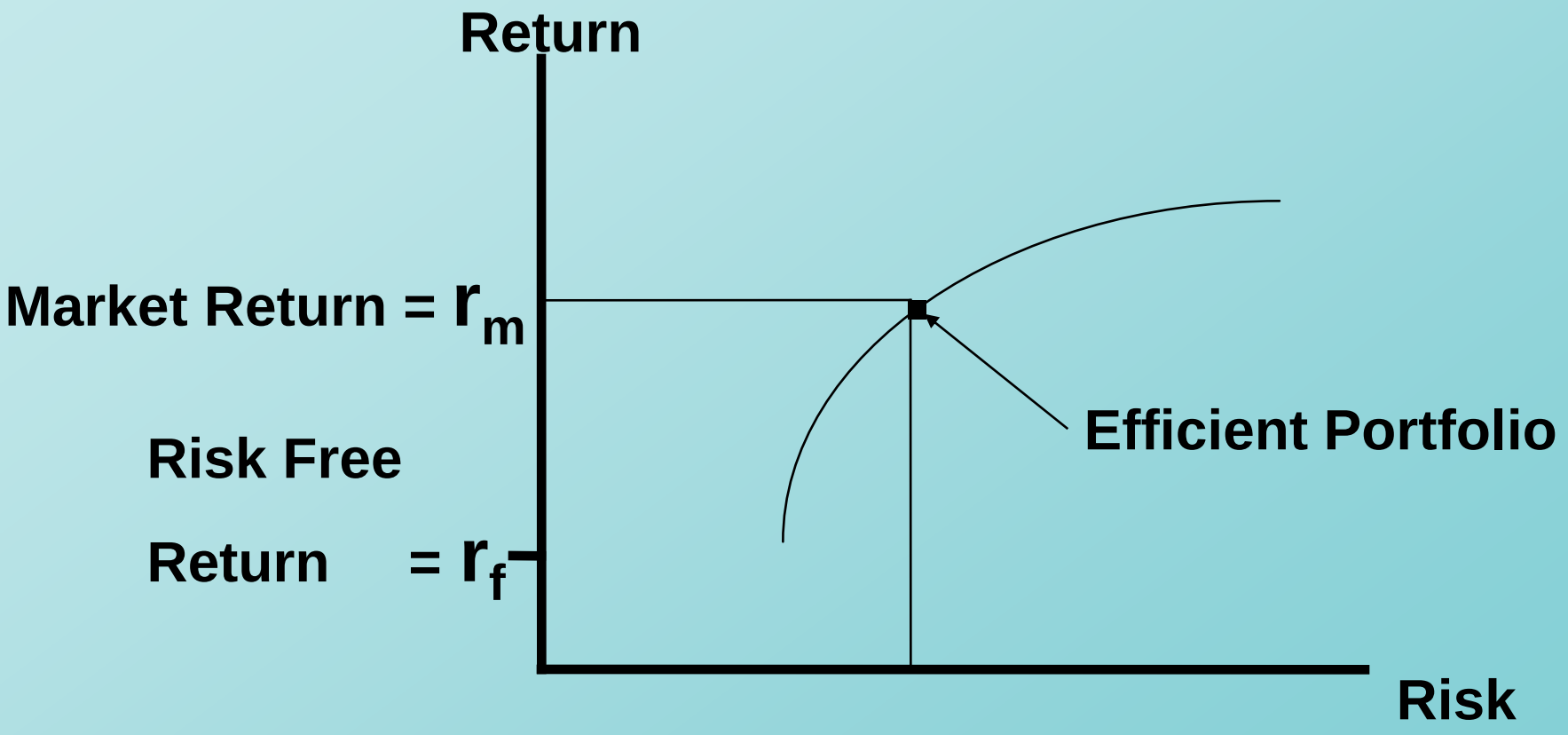
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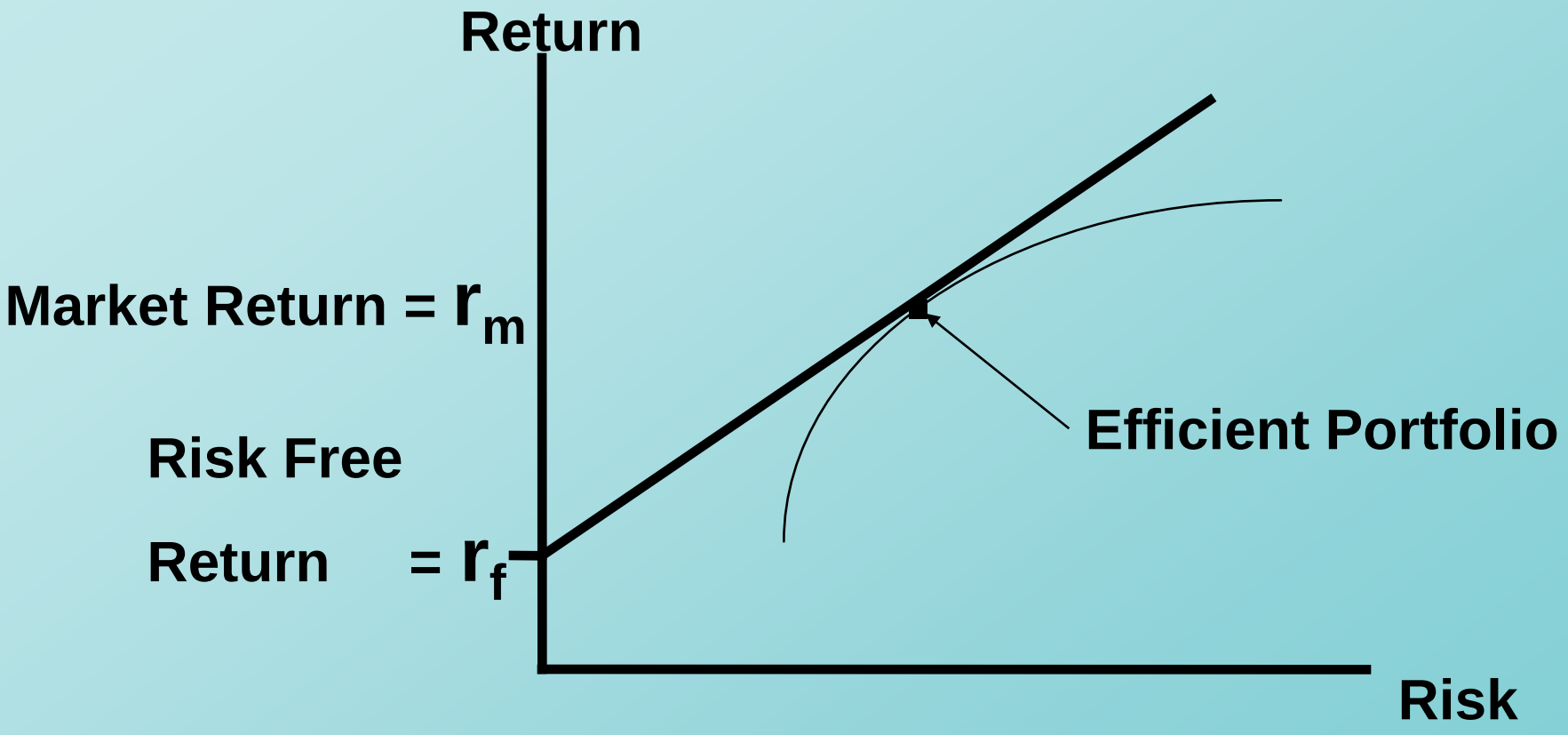
Security Market Line



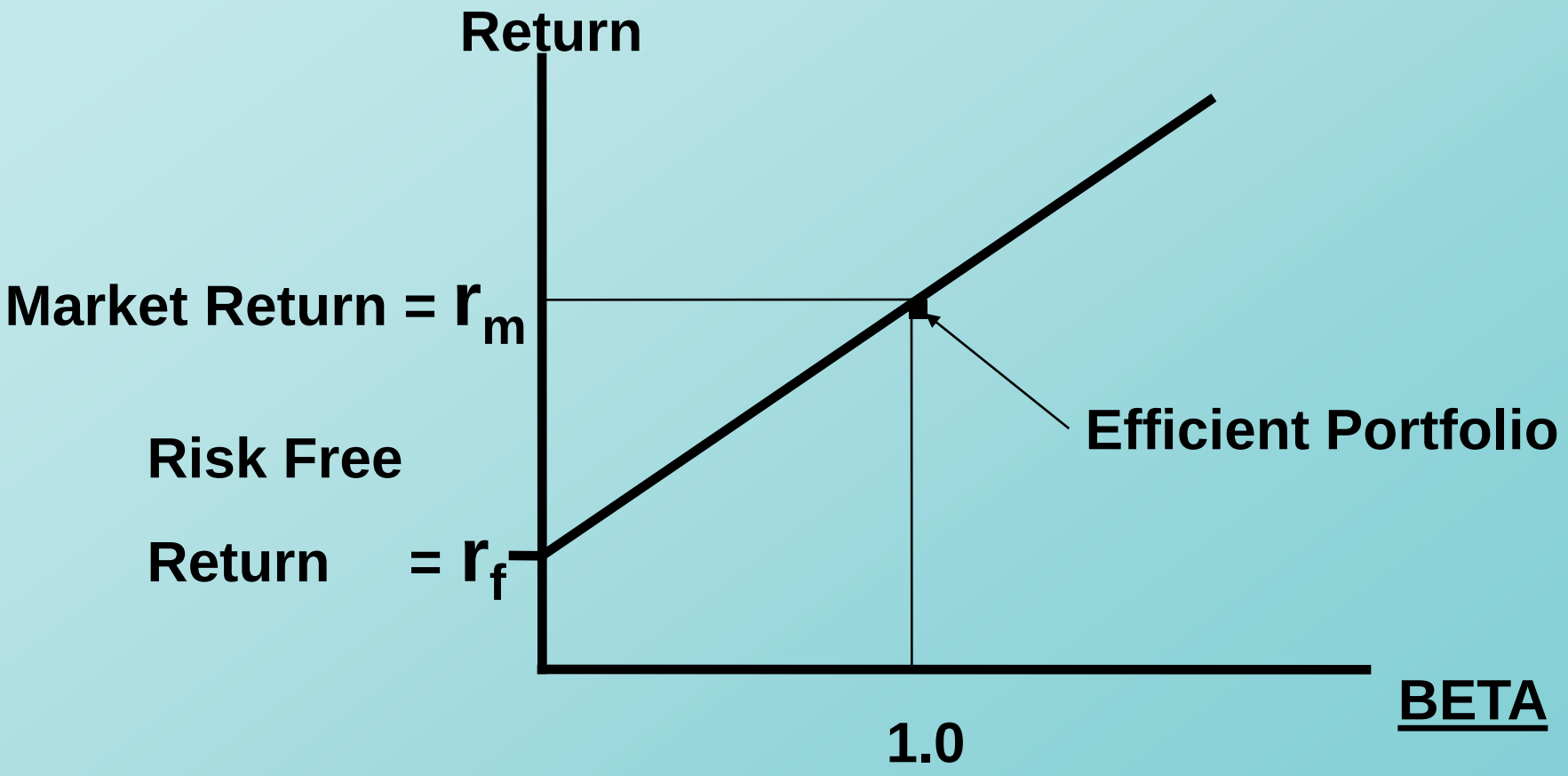
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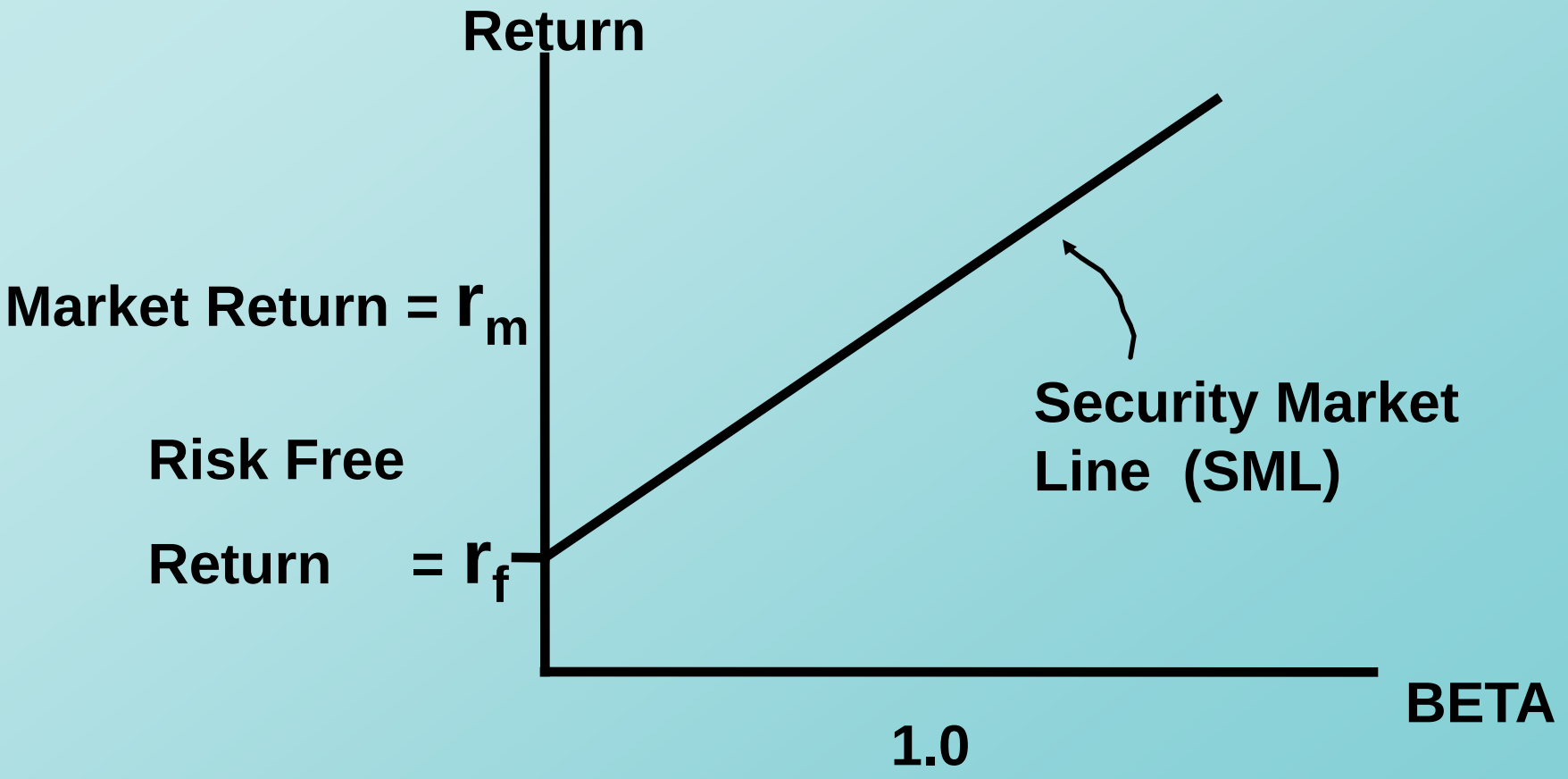
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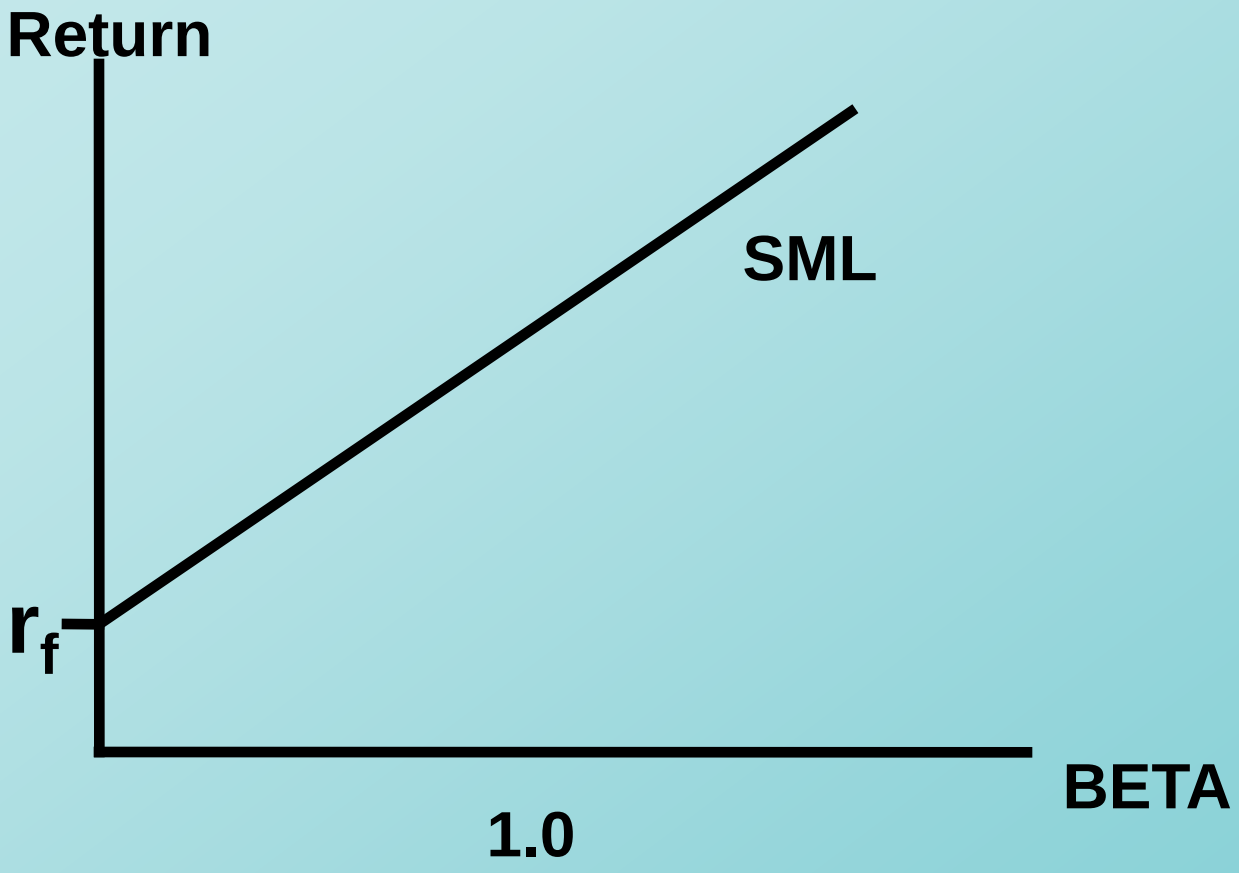
Security Market Line



Security Market Line



Security Market Line



SML Equation = $r_f + B (r_m - r_f)$

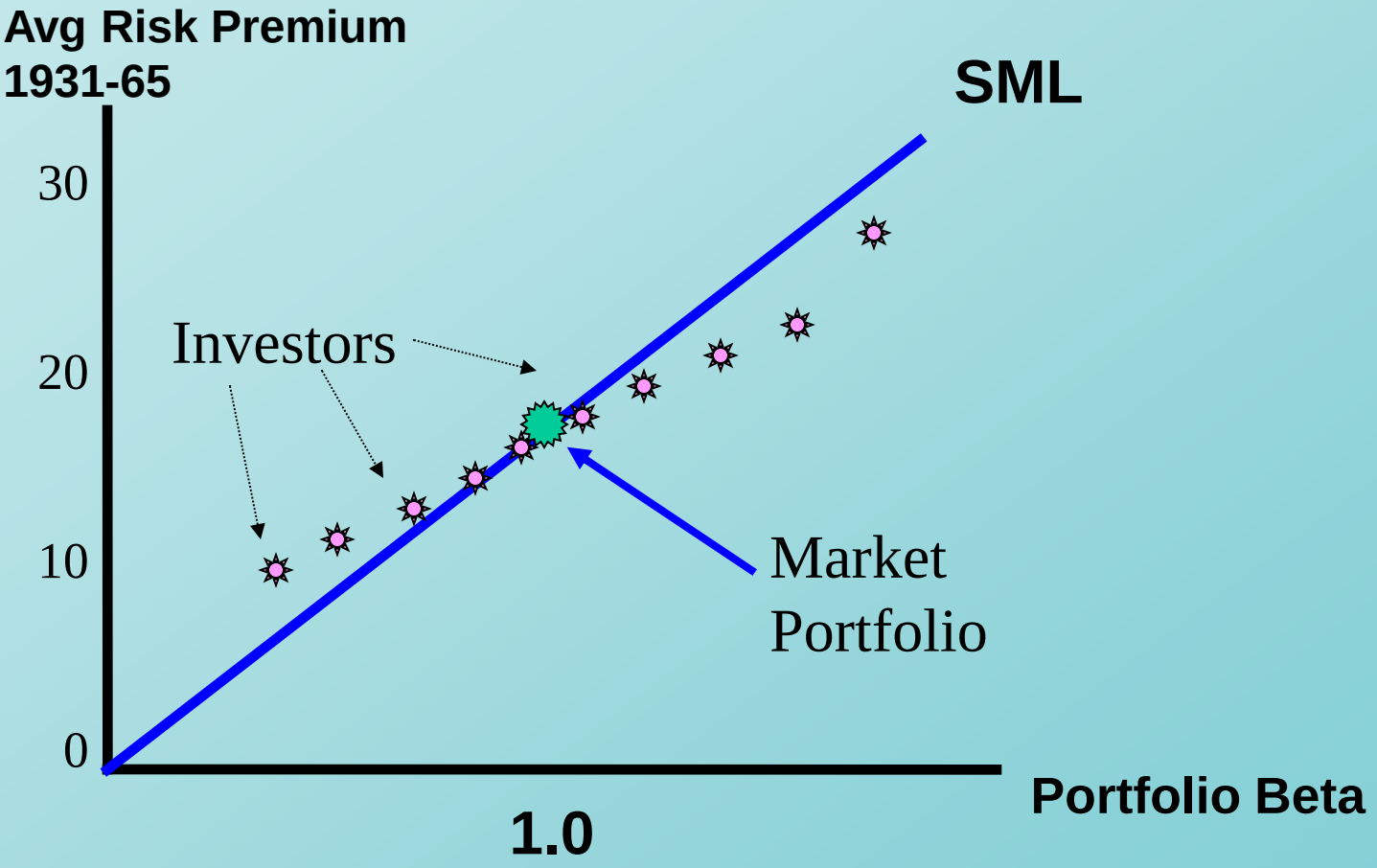
Capital Asset Pricing Model

$$R = r_f + B (r_m - r_f)$$

CAPM

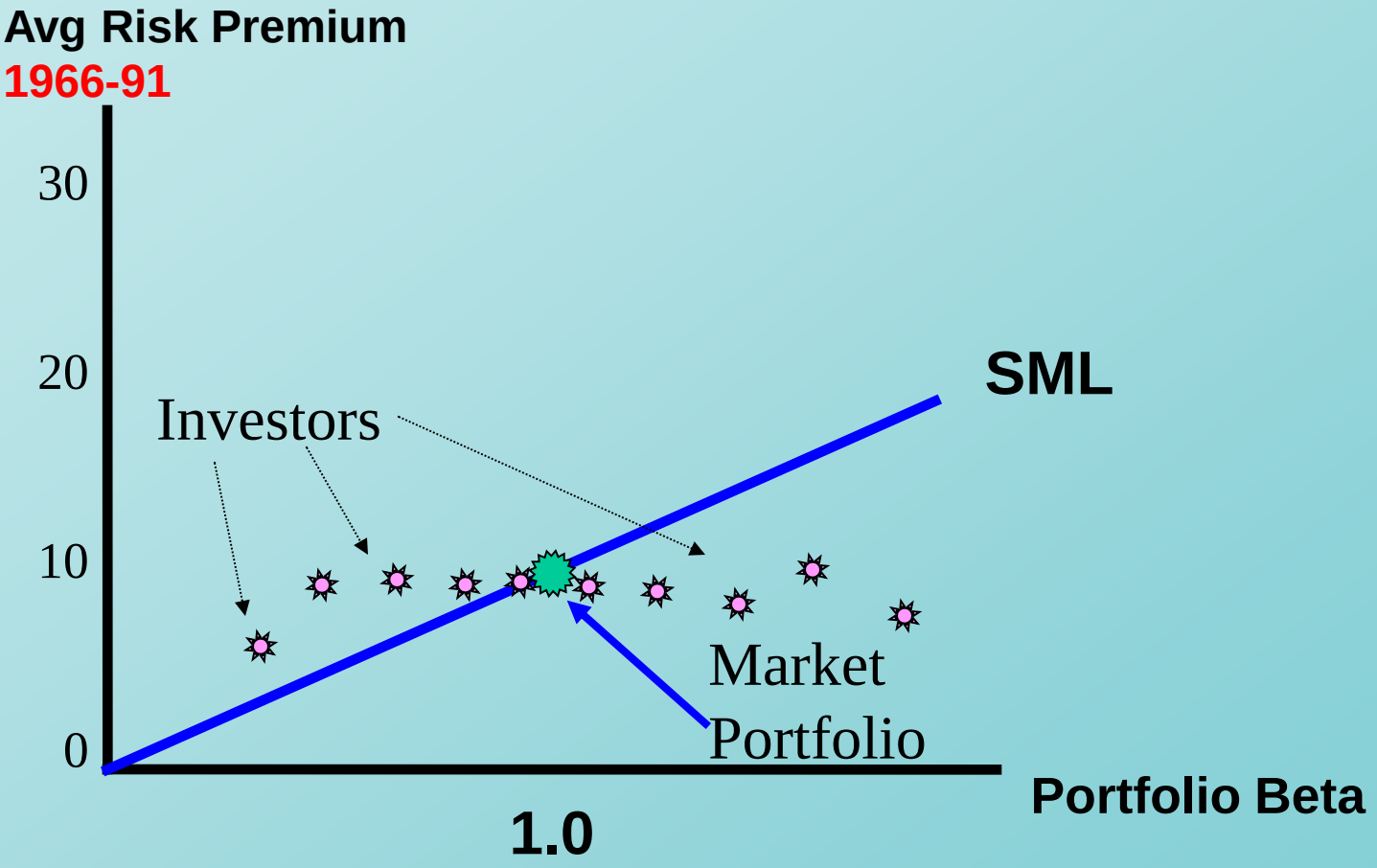
Testing the CAPM

Beta vs. Average Risk Premium



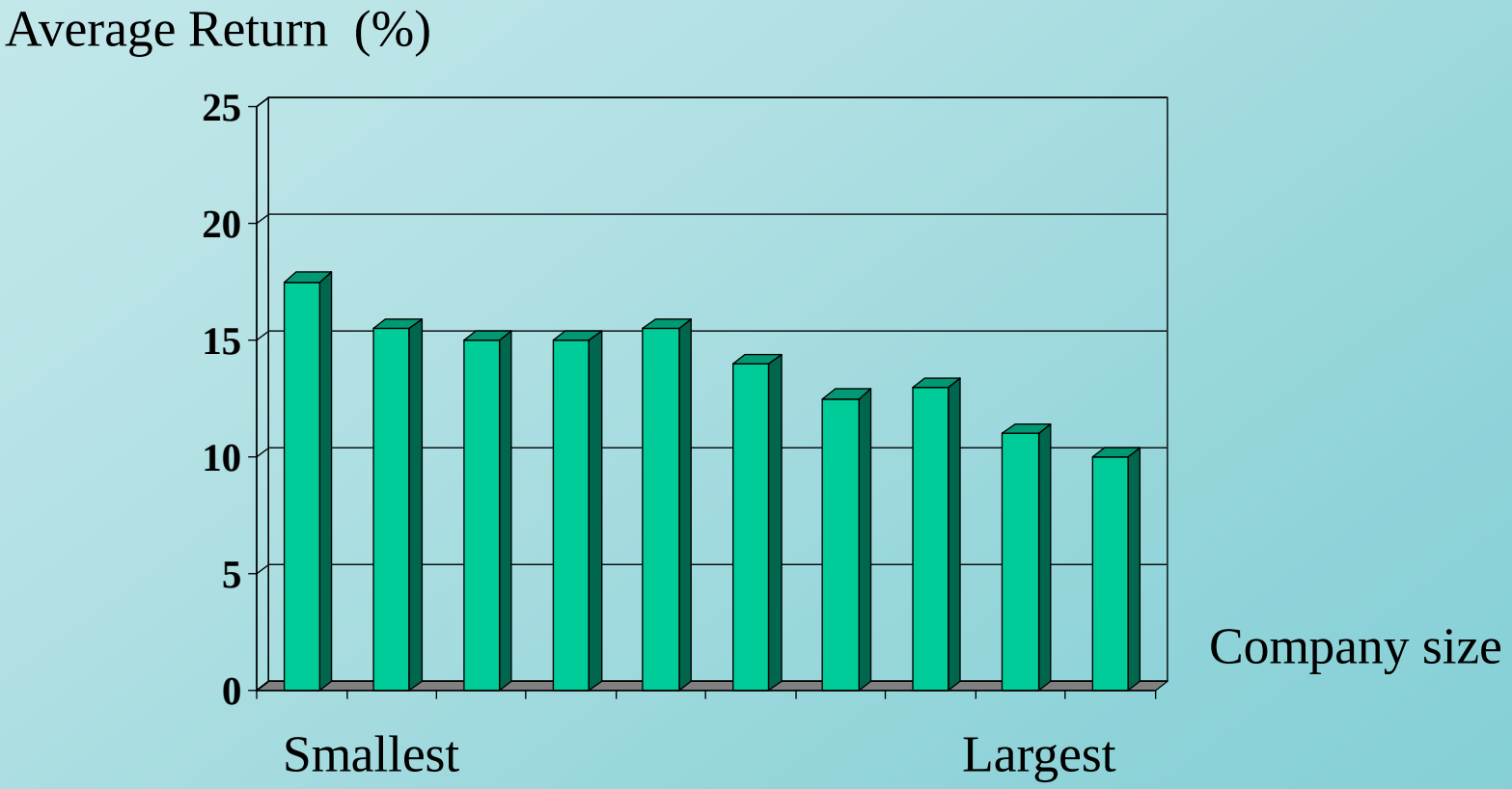
Testing the CAPM

Beta vs. Average Risk Premium



Testing the CAPM

Company Size vs. Average Return



Testing the CAPM

Book-Market vs. Average Return

